

LMM Report – ALM_HT2_harmonised

Generated: 2026-07-08 14:05

Imputations: 10

Random slope: FALSE | Interaction: FALSE

Exposures: 1 | Covariate sets: 2

Model Configuration

Outcome column: ALM_HT2_harmonised
Outcome transform: identity
Estimation method: REML = FALSE (Maximum Likelihood)
Random slope: FALSE
Interaction (exposure × time): FALSE

Exposures:

	exposure	exposure_type	ref_level
1	prot_content_dairy_100g	linear	NA

Covariate sets:

main:
age_at_baseline_scaled, BMI_category, education_level, smoking_status, pa_levels_tertile_f1, diabetes_status, sumtot1_scaled

other_PA:
age_at_baseline_scaled, BMI_category, education_level, smoking_status, pa_levels_who_f1, diabetes_status, sumtot1_scaled

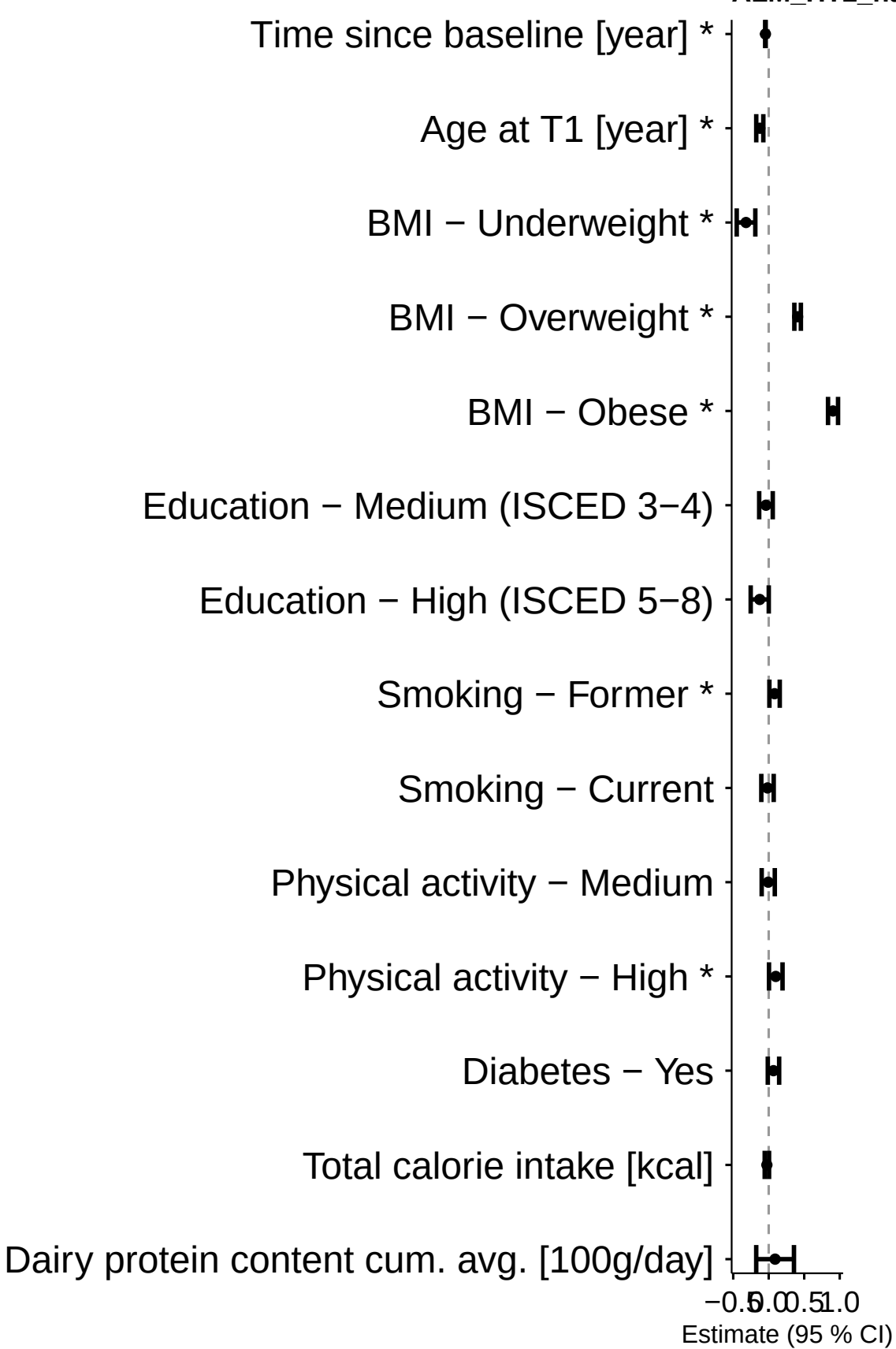
Covariate set: main

age_at_baseline_scaled · BMI_category · education_level · smoking_status · pa_levels_tertile_f1 · diabetes_status · sumtot1_scaled

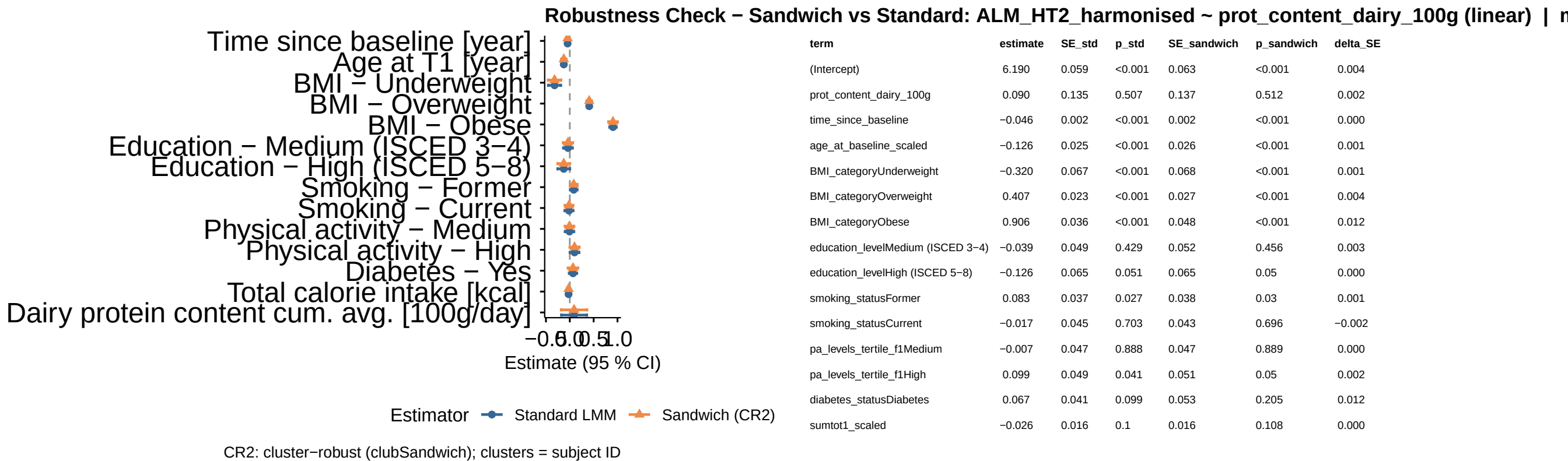
Formula – ALM_HT2_harmonised ~ prot_content_dairy_100g (linear) | main

ALM_HT2_harmonised ~ prot_content_dairy_100g + time_since_baseline +
age_at_baseline_scaled + BMI_category + education_level +
smoking_status + pa_levels_tertile_f1 + diabetes_status +
sumtot1_scaled + (1 | pt)

Imputations pooled: 10



term	estimate	std_error	statistic	df	p_value	conf_low	conf_high
(Intercept)	6.190	0.059	105.751	2375.7	<0.001	6.075	6.305
prot_content_dairy_100g	0.090	0.135	0.664	741.0	0.507	-0.176	0.356
time_since_baseline	-0.046	0.002	-24.231	101.1	<0.001	-0.050	-0.042
age_at_baseline_scaled	-0.126	0.025	-4.950	2865.3	<0.001	-0.175	-0.076
BMI_categoryUnderweight	-0.320	0.067	-4.783	2623.9	<0.001	-0.451	-0.189
BMI_categoryOverweight	0.407	0.023	17.329	568.9	<0.001	0.361	0.453
BMI_categoryObese	0.906	0.036	25.237	543.7	<0.001	0.835	0.976
education_levelMedium (ISCED 3–4)	-0.039	0.049	-0.791	2888.0	0.429	-0.134	0.057
education_levelHigh (ISCED 5–8)	-0.126	0.065	-1.949	2781.4	0.051	-0.254	0.001
smoking_statusFormer	0.083	0.037	2.218	2646.7	0.027	0.010	0.156
smoking_statusCurrent	-0.017	0.045	-0.382	2531.3	0.703	-0.104	0.070
pa_levels_tertile_f1Medium	-0.007	0.047	-0.141	266.8	0.888	-0.099	0.086
pa_levels_tertile_f1High	0.099	0.049	2.046	673.5	0.041	0.004	0.195
diabetes_statusDiabetes	0.067	0.041	1.651	1184.4	0.099	-0.013	0.147
sumtot1_scaled	-0.026	0.016	-1.647	532.4	0.1	-0.056	0.005



Variance Components – ALM_HT2_harmonised ~ prot_content_dairy_100g (linear) | main

```
.. Full VarCorr output .....
Groups   Name          Variance Std.Dev.
pt       (Intercept) 0.313507 0.55992
Residual              0.070473 0.26547

.. Summary .....
Random intercept variance: 0.31351 (SD = 0.55992)
Random slope: not included in this model
Residual variance:         0.07047 (SD = 0.26547)
```

Effect Size (R²) – ALM_HT2_harmonised ~ prot_content_dairy_100g (linear) | main

Marginal R² (fixed effects only): 0.2737

Conditional R² (fixed + random effects): 0.8667

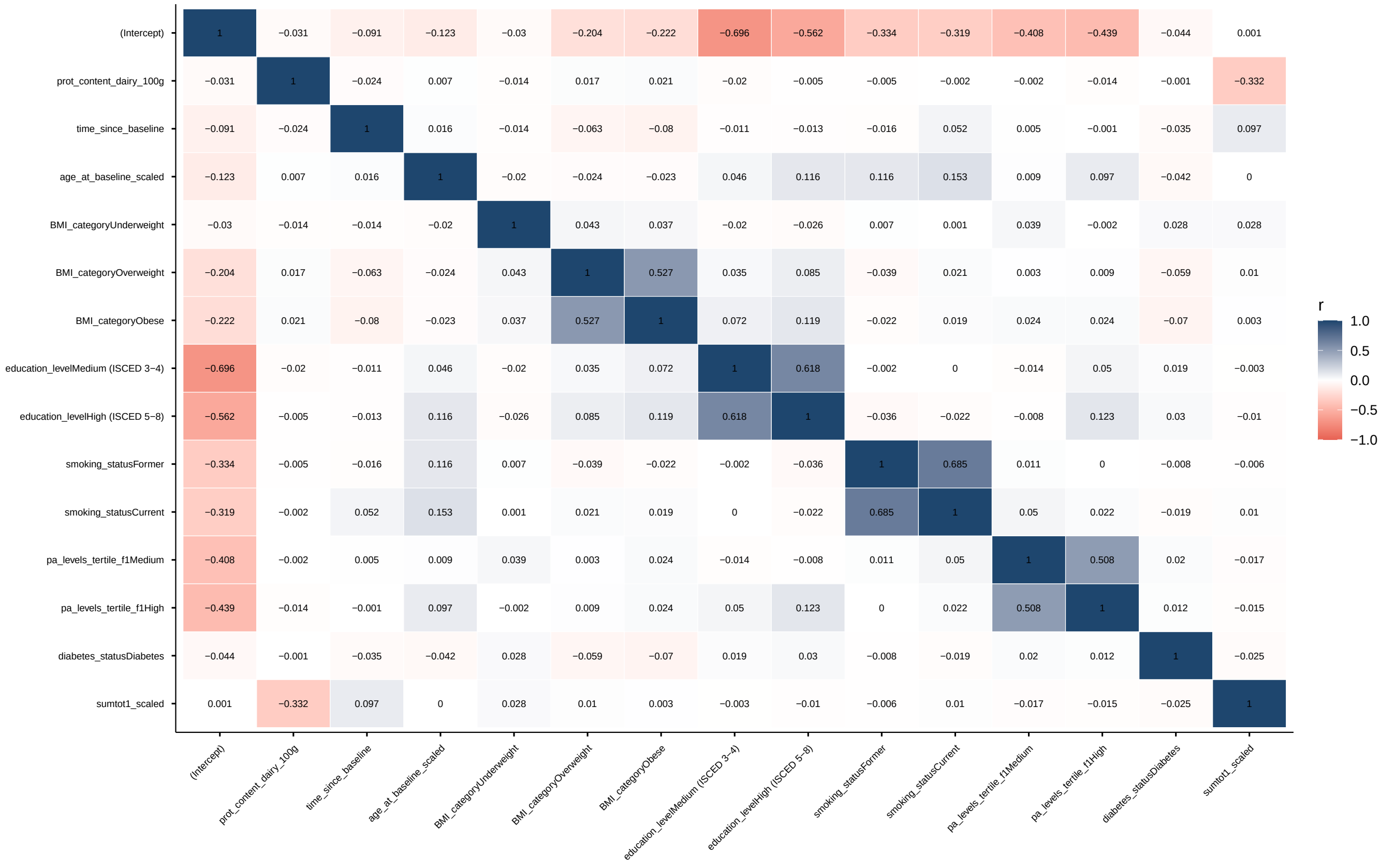
Method: Nakagawa & Schielzeth (2013), Johnson (2014)

VIF – ALM_HT2_harmonised ~ prot_content_dairy_100g (linear) | main

VIF > 5 flagged with '!'

	term	GVIF	Df	$GVIF^{(1/(2*Df))}$	adjusted_vif	flag
	prot_content_dairy_100g	1.126347	1	1.061295	1.061	
	time_since_baseline	1.027155	1	1.013487	1.013	
	age_at_baseline_scaled	1.054438	1	1.026858	1.027	
	BMI_category	1.046307	3	1.007573	1.008	
	education_level	1.059187	2	1.014479	1.014	
	smoking_status	1.047092	2	1.011571	1.012	
	pa_levels_tertile_f1	1.039800	2	1.009805	1.010	
	diabetes_status	1.013296	1	1.006626	1.007	
	sumtot1_scaled	1.137033	1	1.066318	1.066	

Fixed-Effect Correlations – ALM_HT2_harmonised ~ prot_content_dairy_100g (linear) | main



Influential Points – ALM_HT2_harmonised ~ prot_content_dairy_100g (linear) | main

Influential observations (top 1% Cook's D > 0.29635): 30

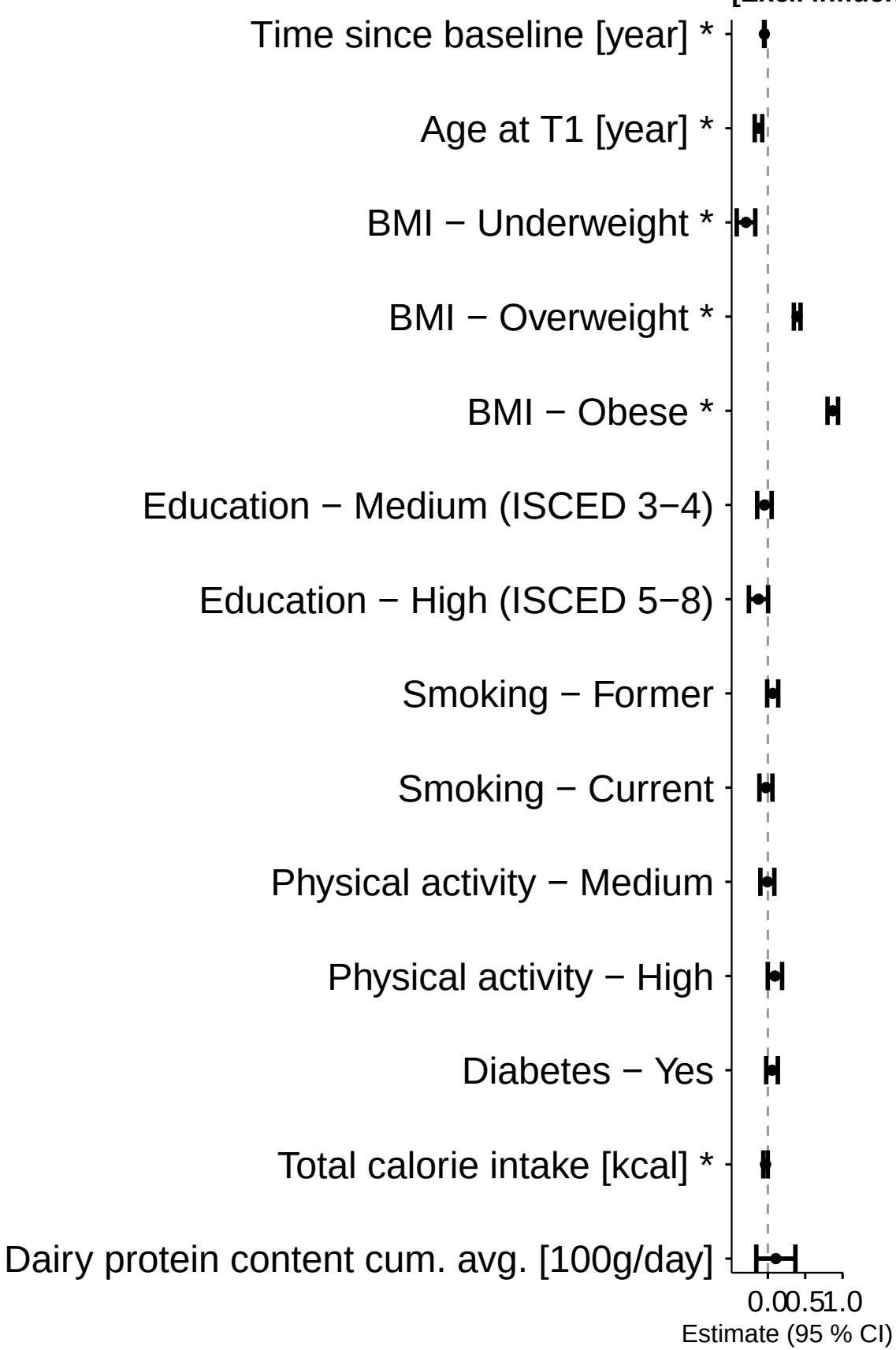
Index	Cook's D
index	cooks_d
172	0.29727
196	0.33485
352	0.34436
353	0.51174
357	0.41628
532	0.30554
538	0.29895
724	0.36483
927	0.30695
1061	0.37902
1152	0.54642
1182	0.38205
1218	0.33663
1338	0.32262
1433	0.32453
1544	0.50078
1545	0.79852
1715	0.31390
1717	0.37771
1778	0.47919
1785	0.29964
1864	0.48776
2193	0.47534
2194	1.16175
2251	0.55916
2316	0.47789
2317	0.43733
2625	0.35780
2631	0.43177
2632	0.60096

Formula – [Excl. influential (top 1% Cook's D)] ALM_HT2_harmonised ~ prot_content_dairy_100g (linear) | main

Excluded participants (top 1% Cook's D, imp 1): 24

Remaining n: 1042

ALM_HT2_harmonised ~ prot_content_dairy_100g + time_since_baseline +
age_at_baseline_scaled + BMI_category + education_level +
smoking_status + pa_levels_tertile_f1 + diabetes_status +
sumtot1_scaled + (1 | pt)

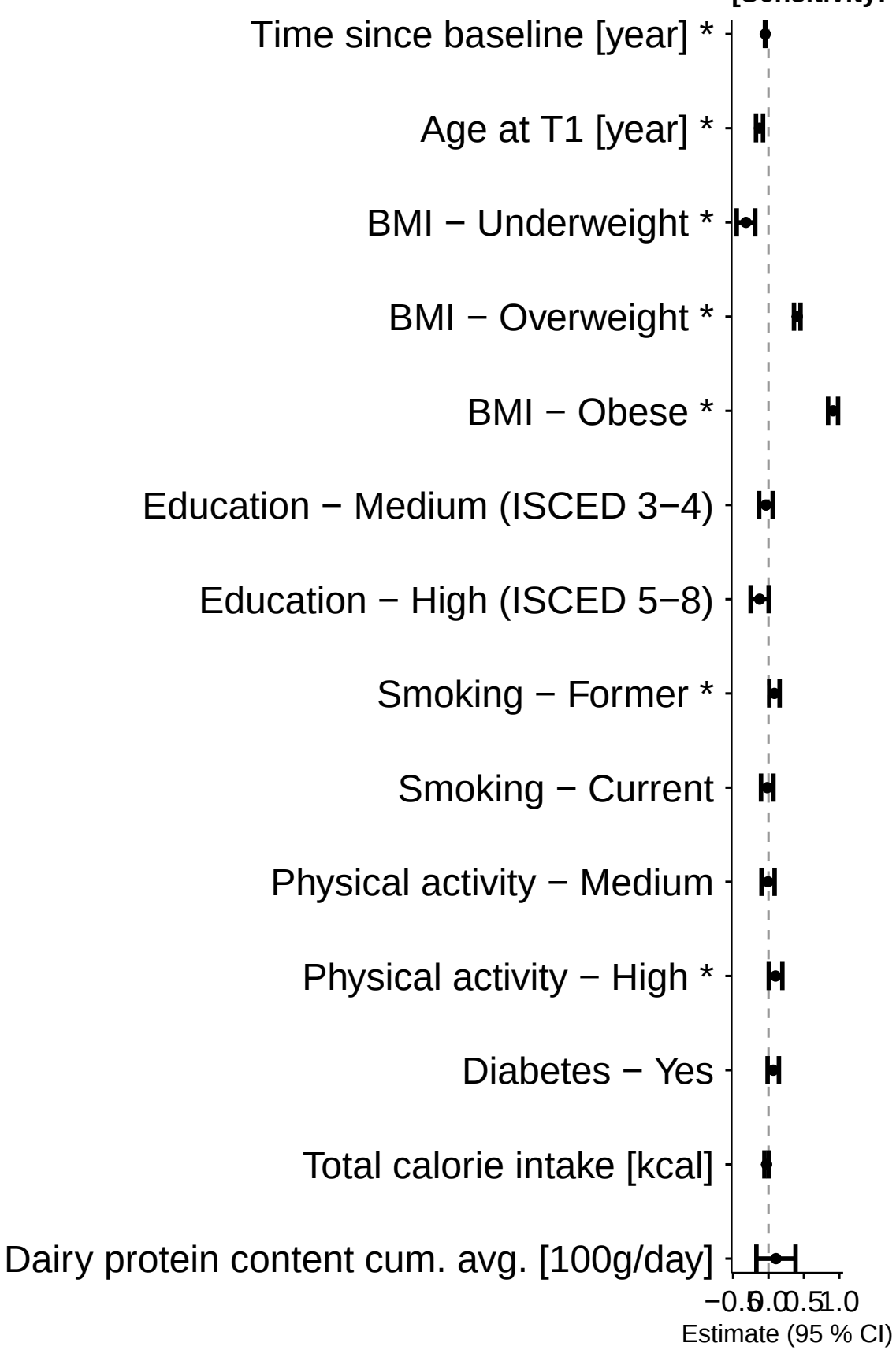


term	estimate	std_error	statistic	df	p_value	conf_low	conf_high
(Intercept)	6.205	0.059	105.197	2363.8	<0.001	6.090	6.321
prot_content_dairy_100g	0.107	0.133	0.803	488.7	0.422	-0.155	0.369
time_since_baseline	-0.047	0.002	-25.190	74.9	<0.001	-0.050	-0.043
age_at_baseline_scaled	-0.126	0.026	-4.907	2811.7	<0.001	-0.176	-0.075
BMI_categoryUnderweight	-0.293	0.063	-4.622	2330.7	<0.001	-0.417	-0.169
BMI_categoryOverweight	0.393	0.023	17.258	419.7	<0.001	0.348	0.438
BMI_categoryObese	0.869	0.036	24.255	326.5	<0.001	0.799	0.940
education_levelMedium (ISCED 3-4)	-0.046	0.049	-0.931	2838.2	0.352	-0.143	0.051
education_levelHigh (ISCED 5-8)	-0.125	0.065	-1.911	2732.8	0.056	-0.254	0.003
smoking_statusFormer	0.065	0.038	1.721	2467.7	0.085	-0.009	0.139
smoking_statusCurrent	-0.026	0.045	-0.579	2249.7	0.563	-0.113	0.062
pa_levels_tertile_f1Medium	-0.007	0.047	-0.144	274.3	0.886	-0.100	0.087
pa_levels_tertile_f1High	0.095	0.049	1.944	846.1	0.052	-0.001	0.190
diabetes_statusDiabetes	0.056	0.040	1.404	967.1	0.161	-0.022	0.134
sumtot1_scaled	-0.031	0.015	-2.037	254.1	0.043	-0.061	-0.001

Formula – [Sensitivity: dairy <= 1000 g/day] ALM_HT2_harmonised ~ prot_content_dairy_100g (linear) | main

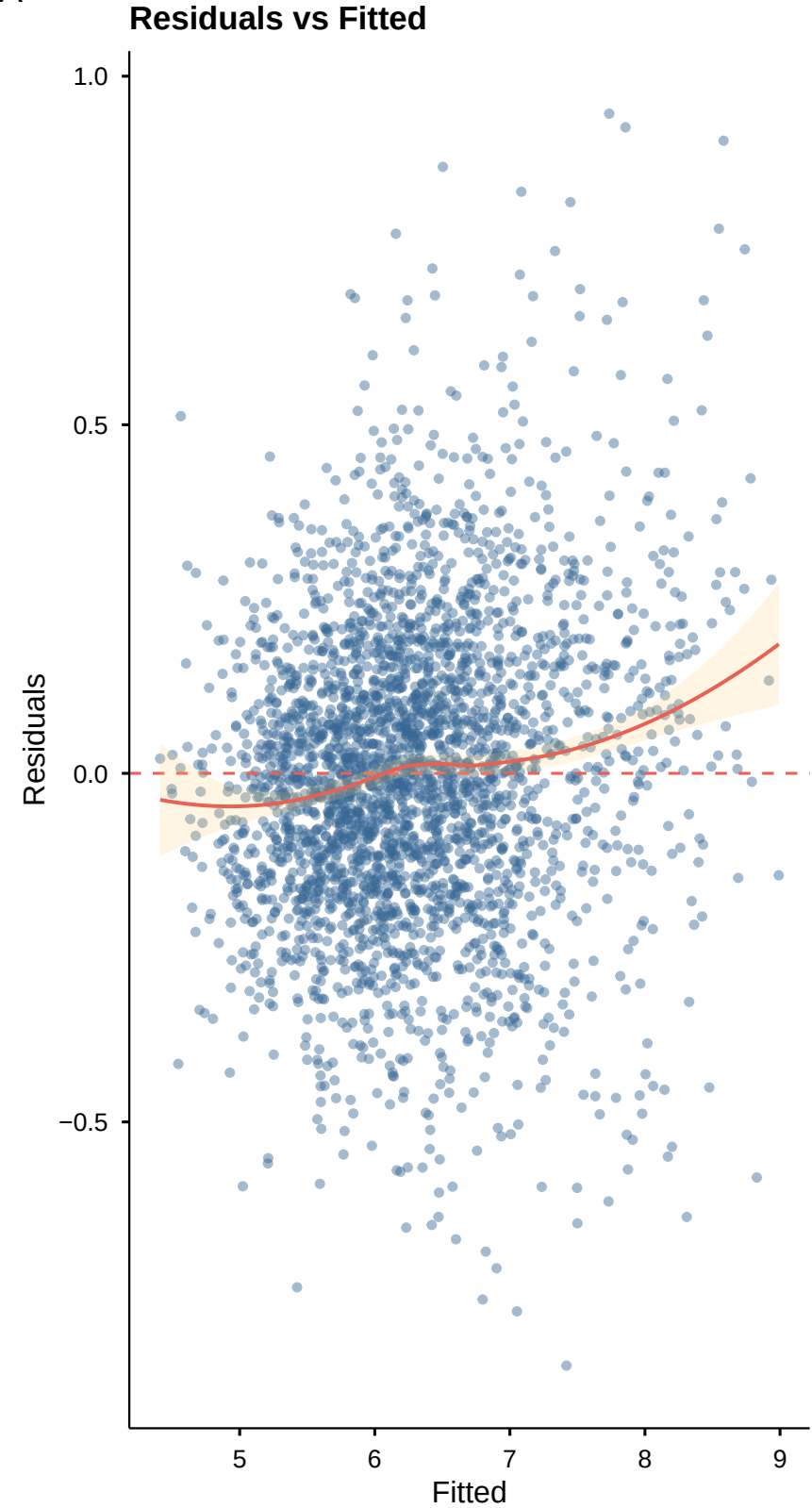
ALM_HT2_harmonised ~ prot_content_dairy_100g + time_since_baseline +
age_at_baseline_scaled + BMI_category + education_level +
smoking_status + pa_levels_tertile_f1 + diabetes_status +
sumtot1_scaled + (1 | pt)

Imputations pooled: 10
Participants in sensitivity sample: 1061
Threshold: dairy_total_gday_cumavg <= 1000 g/day

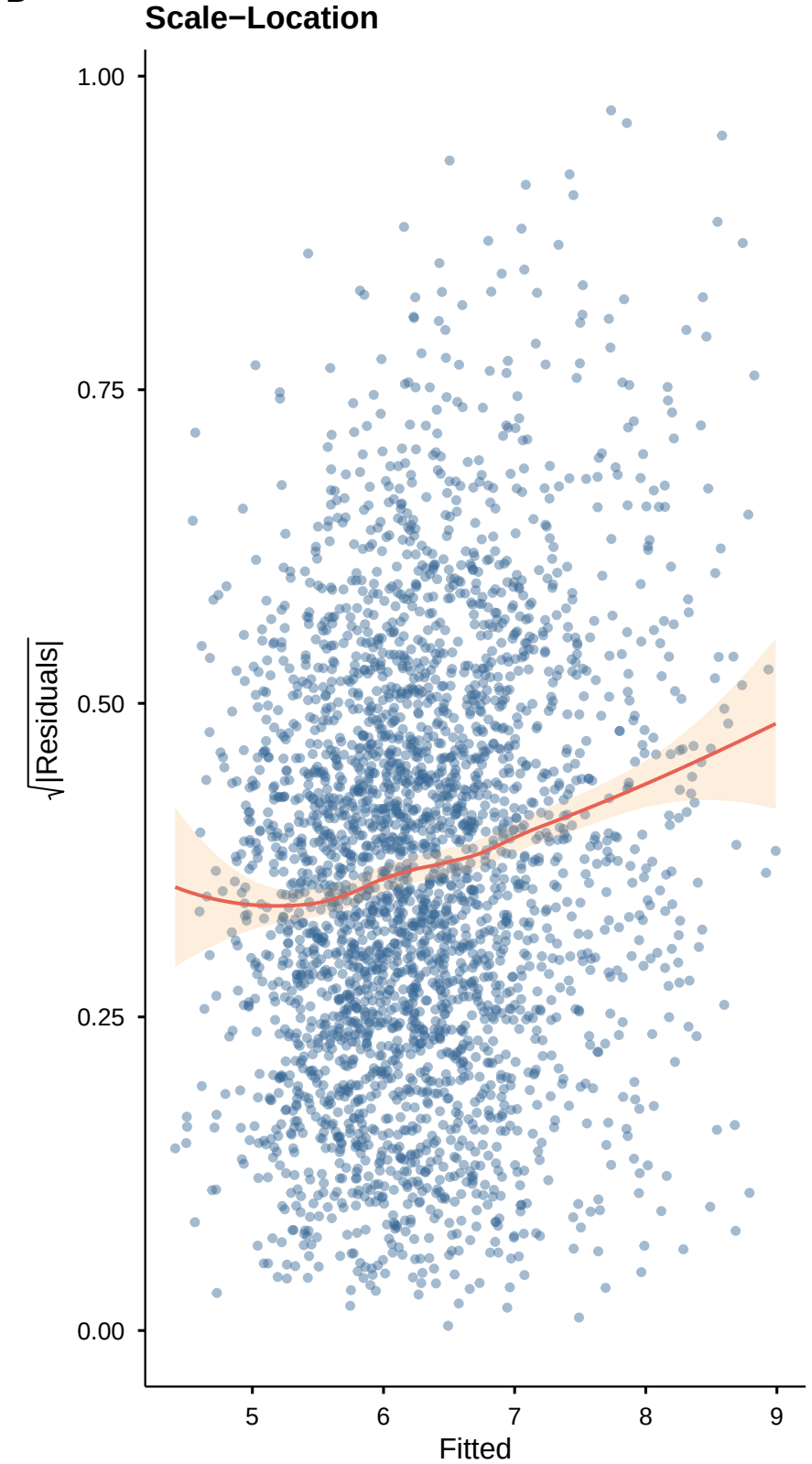


term	estimate	std_error	statistic	df	p_value	conf_low	conf_high
(Intercept)	6.189	0.059	105.536	2440.5	<0.001	6.074	6.303
prot_content_dairy_100g	0.105	0.141	0.748	629.6	0.455	-0.171	0.382
time_since_baseline	-0.046	0.002	-24.171	94.3	<0.001	-0.050	-0.043
age_at_baseline_scaled	-0.127	0.025	-5.008	2852.5	<0.001	-0.177	-0.077
BMI_categoryUnderweight	-0.319	0.067	-4.779	2629.8	<0.001	-0.449	-0.188
BMI_categoryOverweight	0.406	0.023	17.281	591.9	<0.001	0.360	0.452
BMI_categoryObese	0.912	0.036	25.464	611.2	<0.001	0.842	0.982
education_levelMedium (ISCED 3-4)	-0.037	0.049	-0.750	2882.5	0.453	-0.132	0.059
education_levelHigh (ISCED 5-8)	-0.125	0.065	-1.926	2771.8	0.054	-0.253	0.002
smoking_statusFormer	0.084	0.037	2.241	2631.3	0.025	0.010	0.157
smoking_statusCurrent	-0.018	0.045	-0.410	2423.8	0.682	-0.106	0.069
pa_levels_tertile_f1Medium	-0.006	0.047	-0.138	291.3	0.89	-0.099	0.086
pa_levels_tertile_f1High	0.100	0.049	2.045	643.1	0.041	0.004	0.195
diabetes_statusDiabetes	0.067	0.041	1.644	1169.0	0.1	-0.013	0.147
sumtot1_scaled	-0.027	0.015	-1.768	677.9	0.077	-0.058	0.003

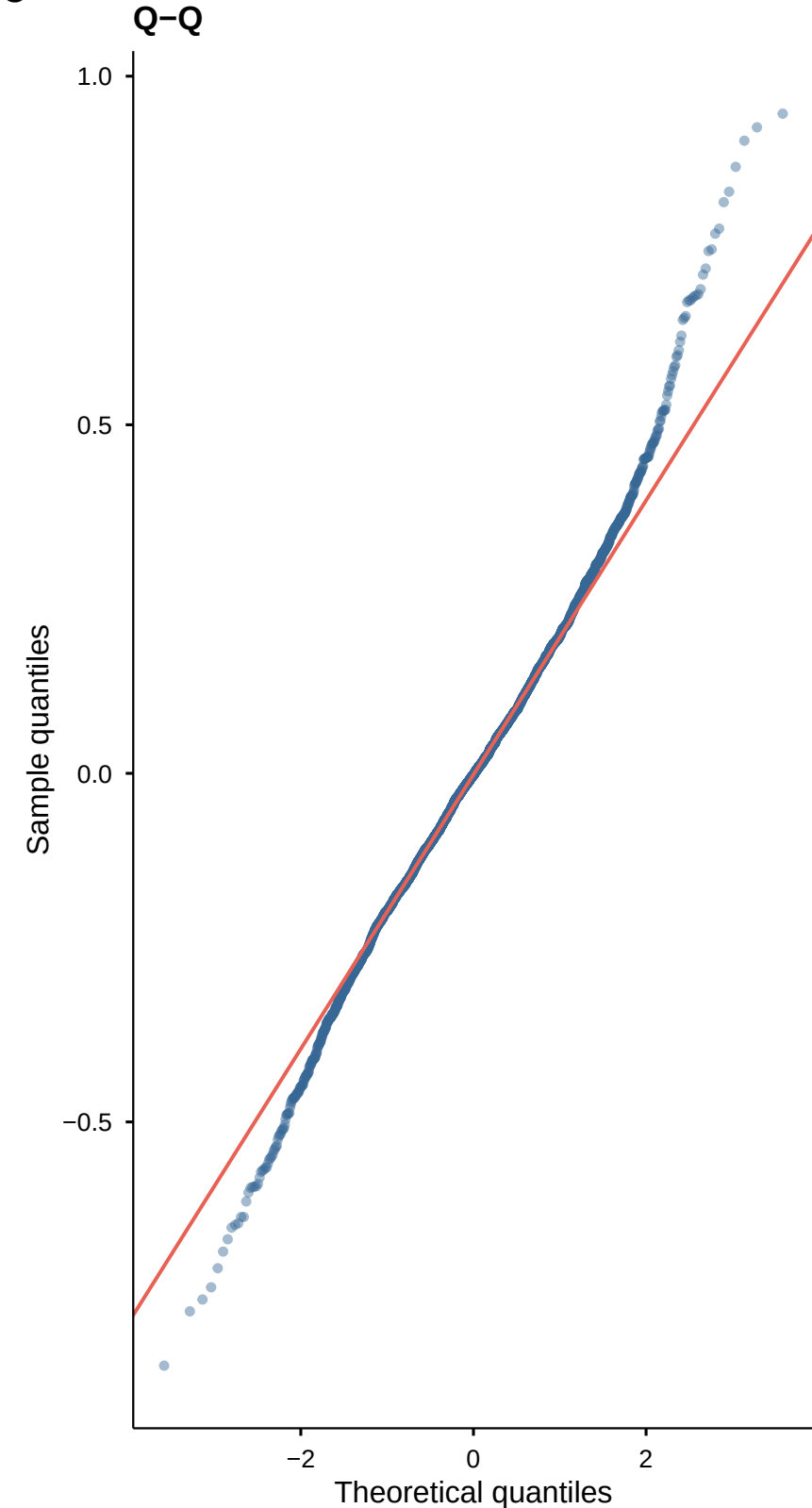
A



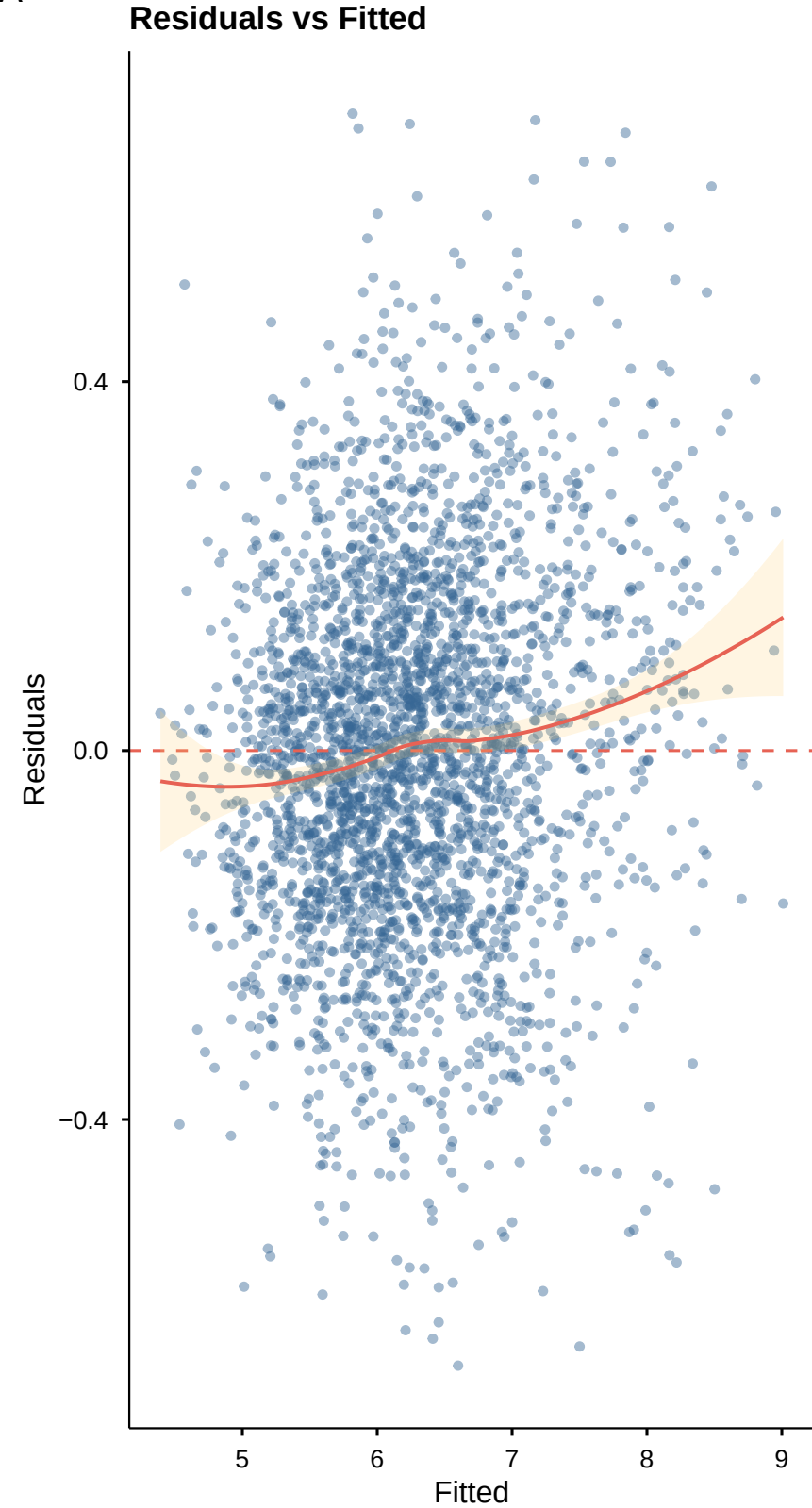
B



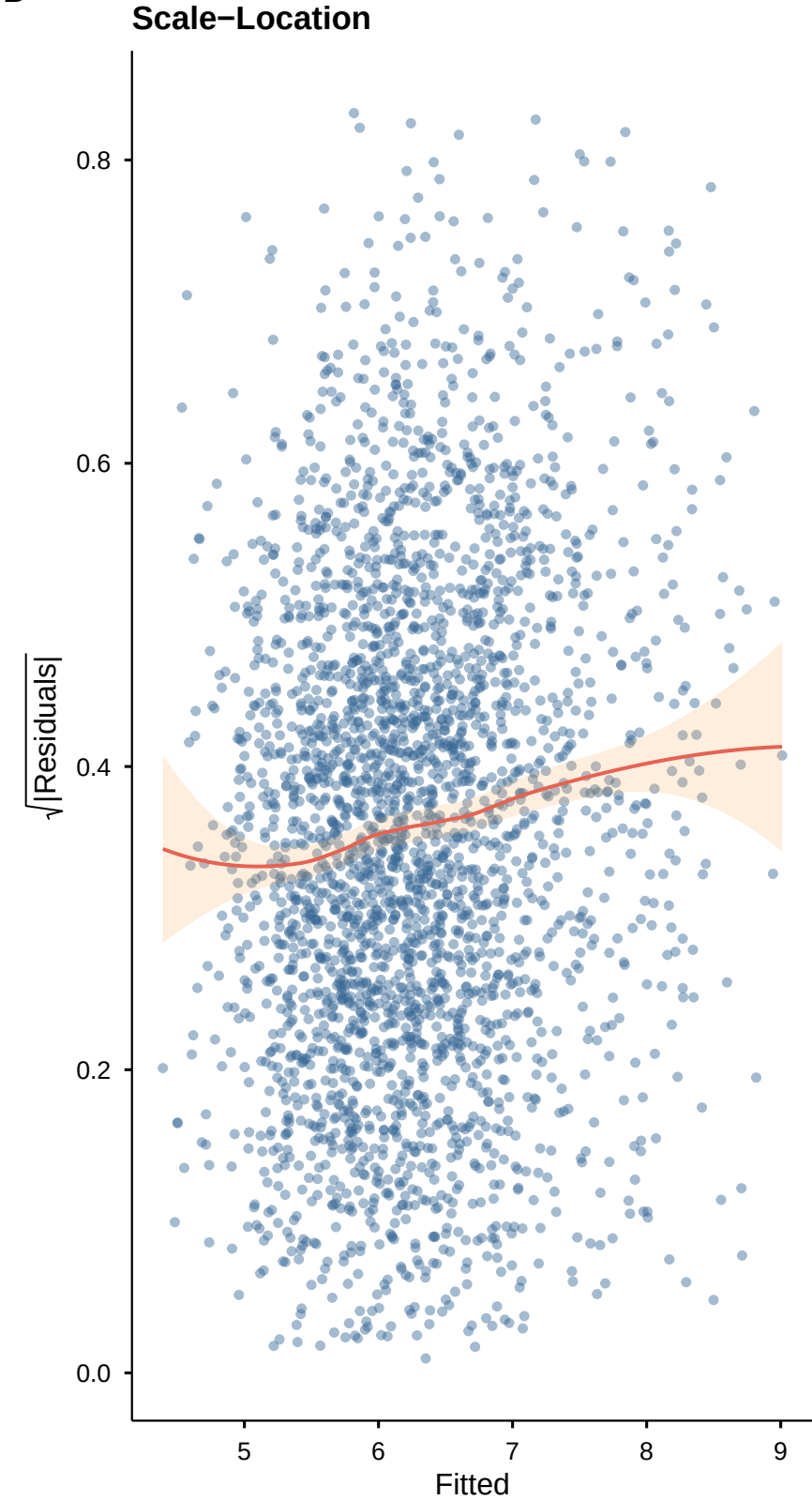
C



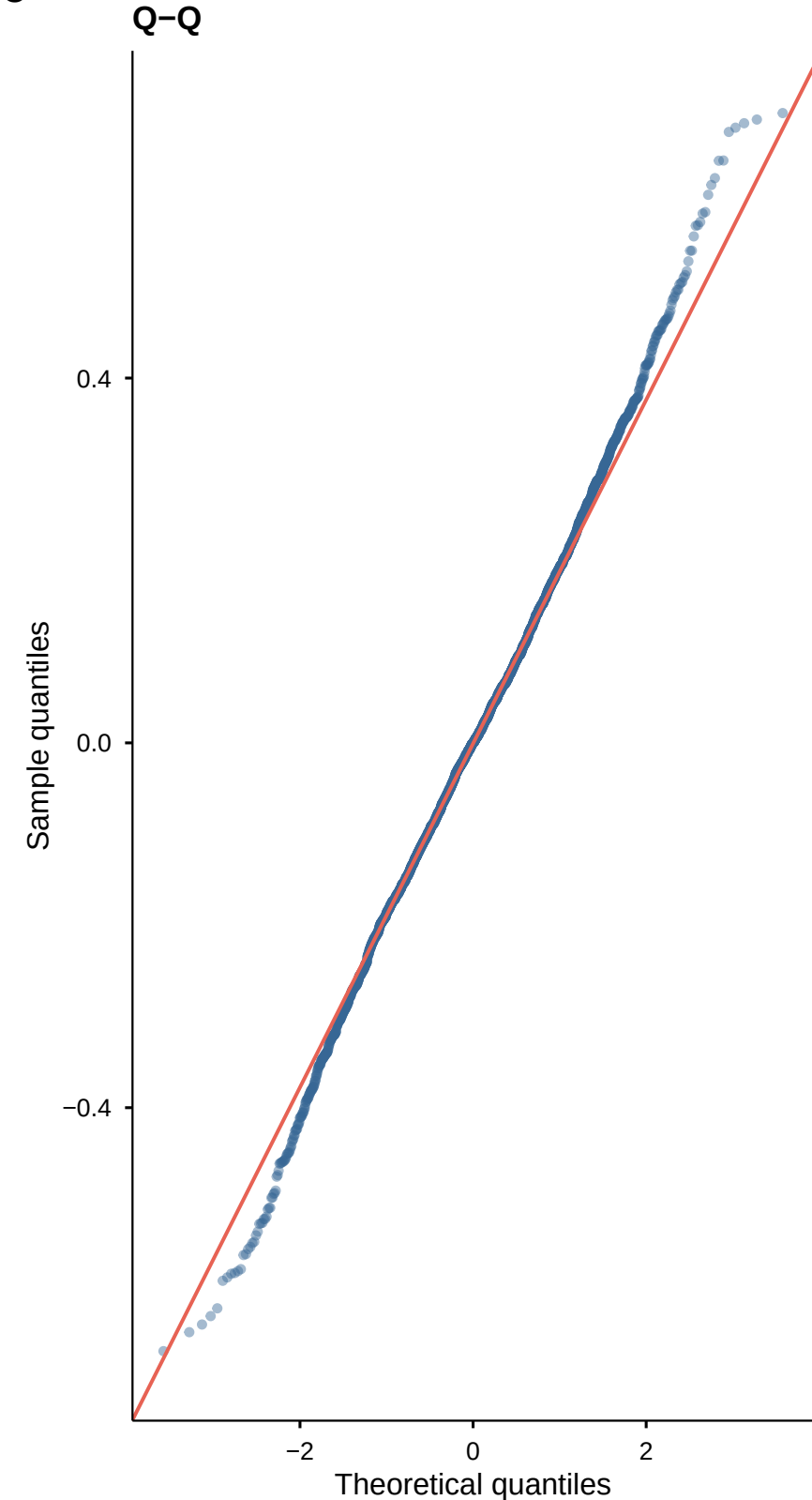
A



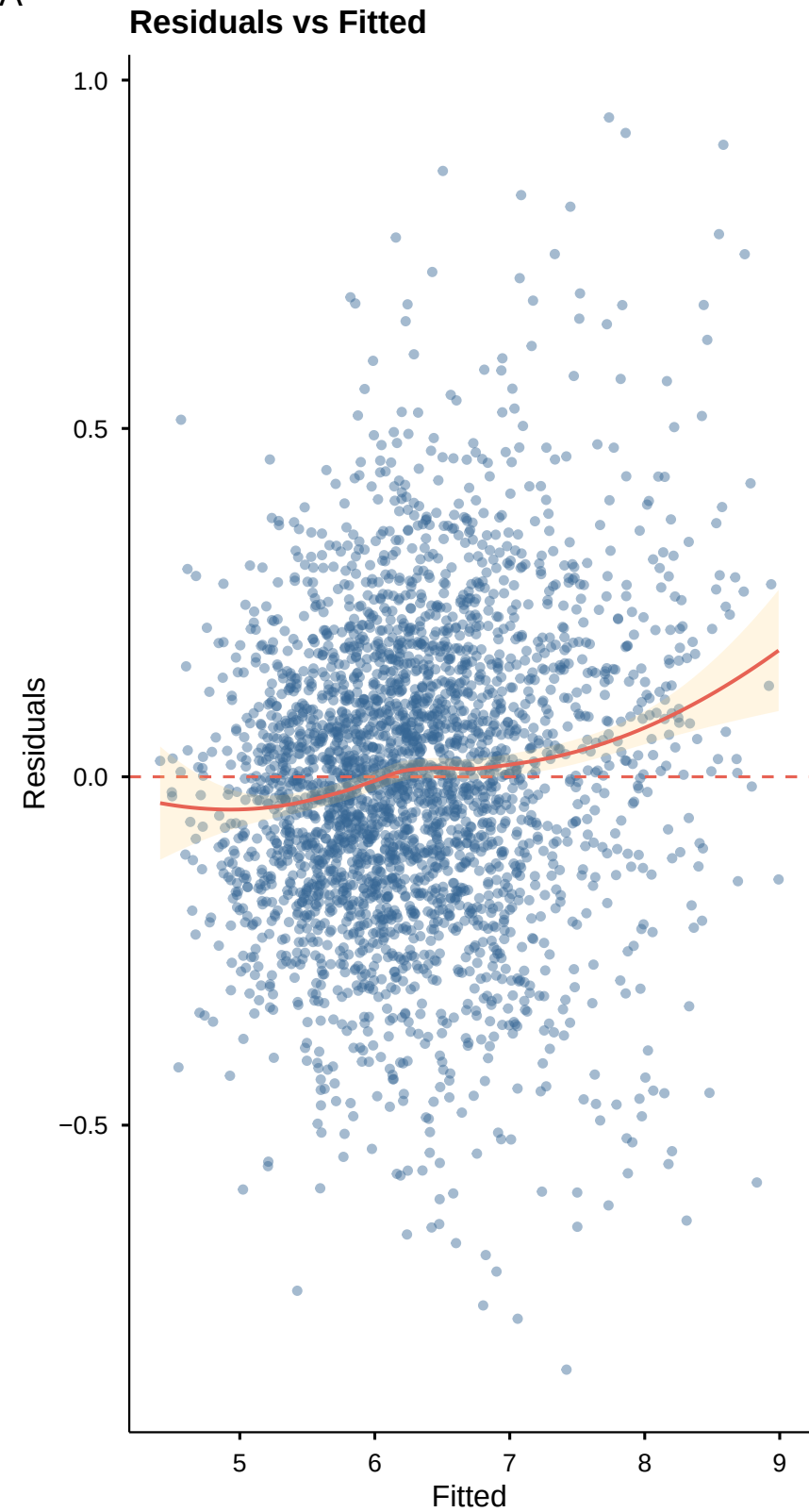
B



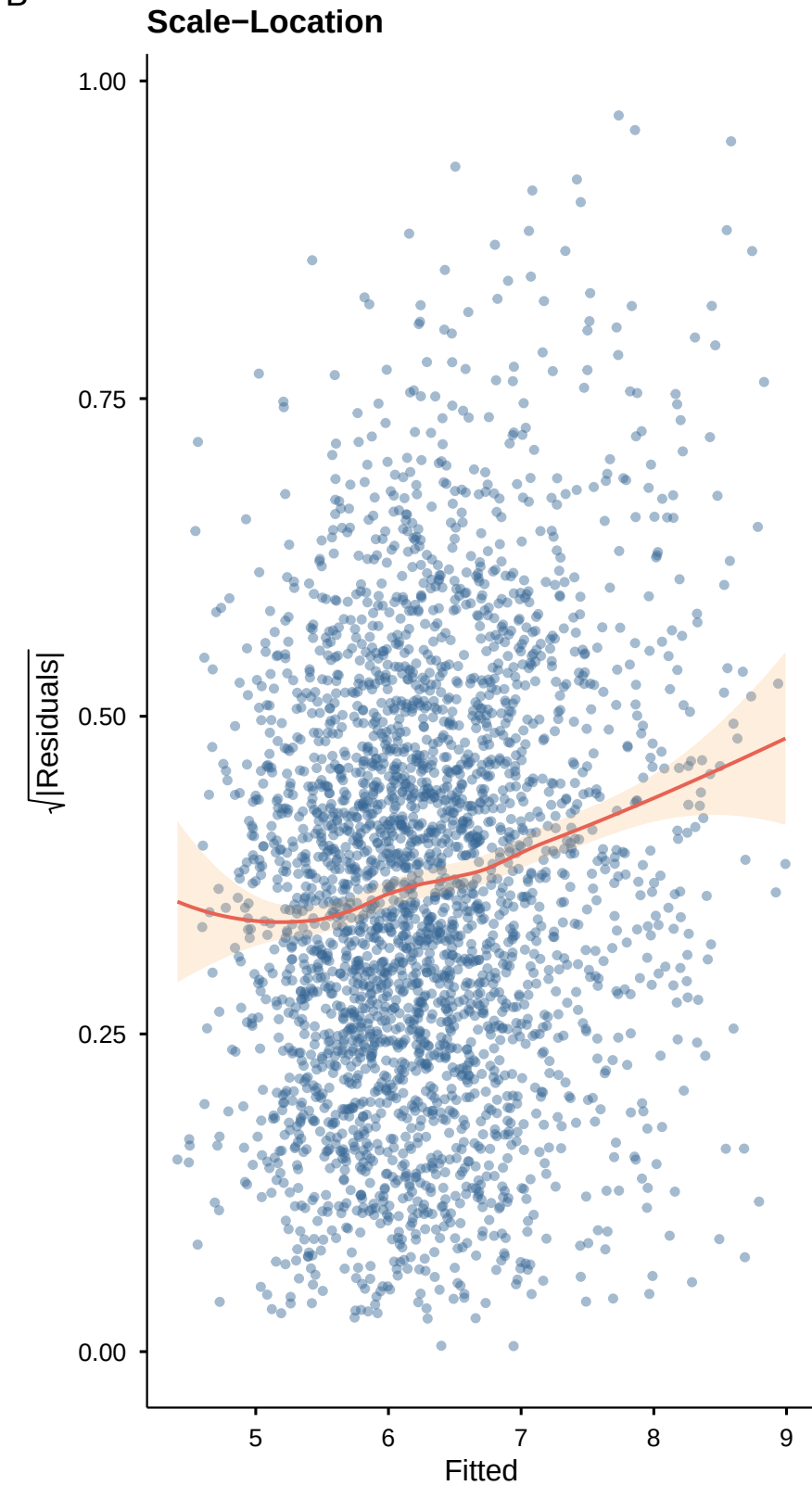
C



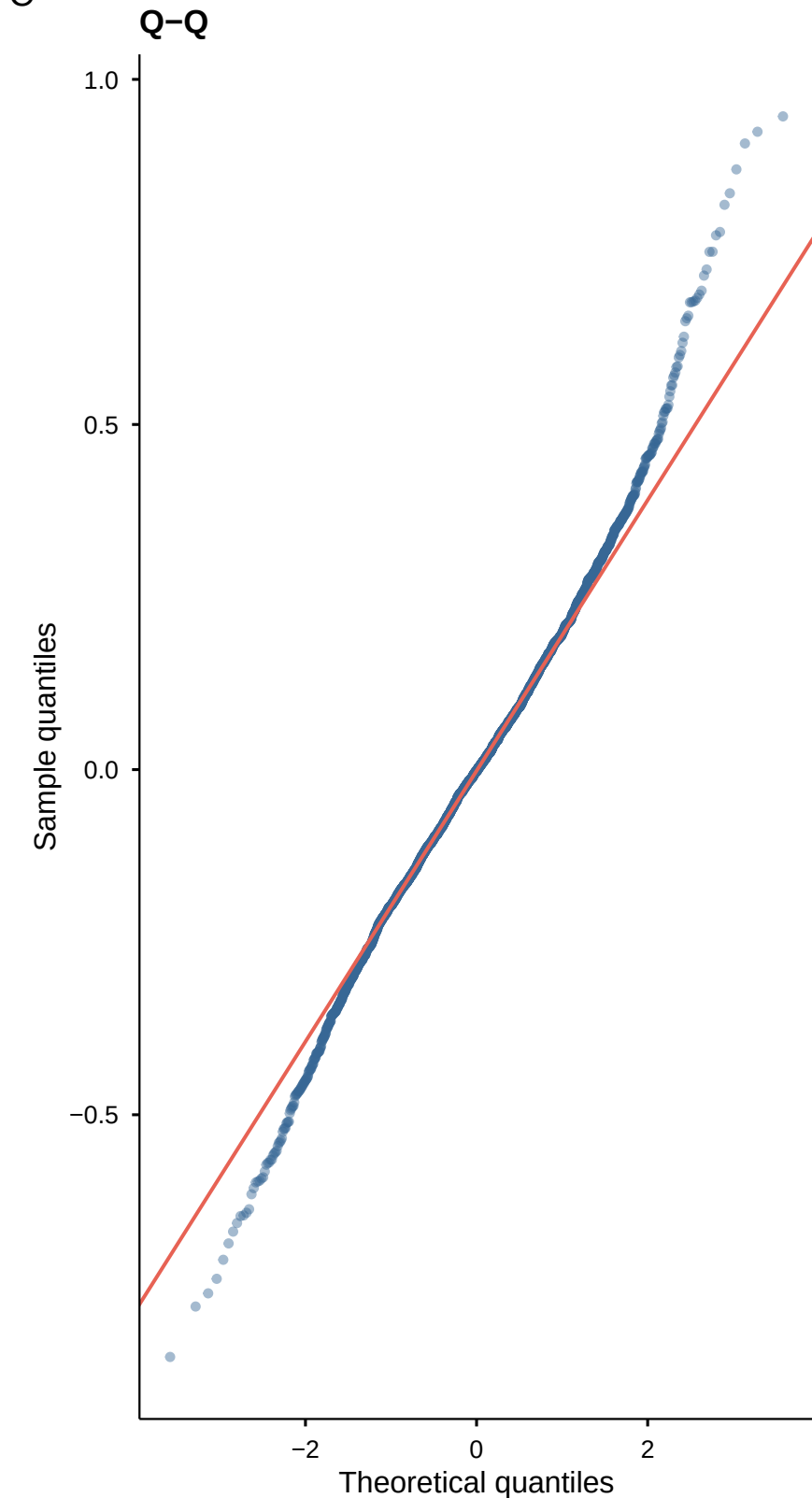
A



B



C



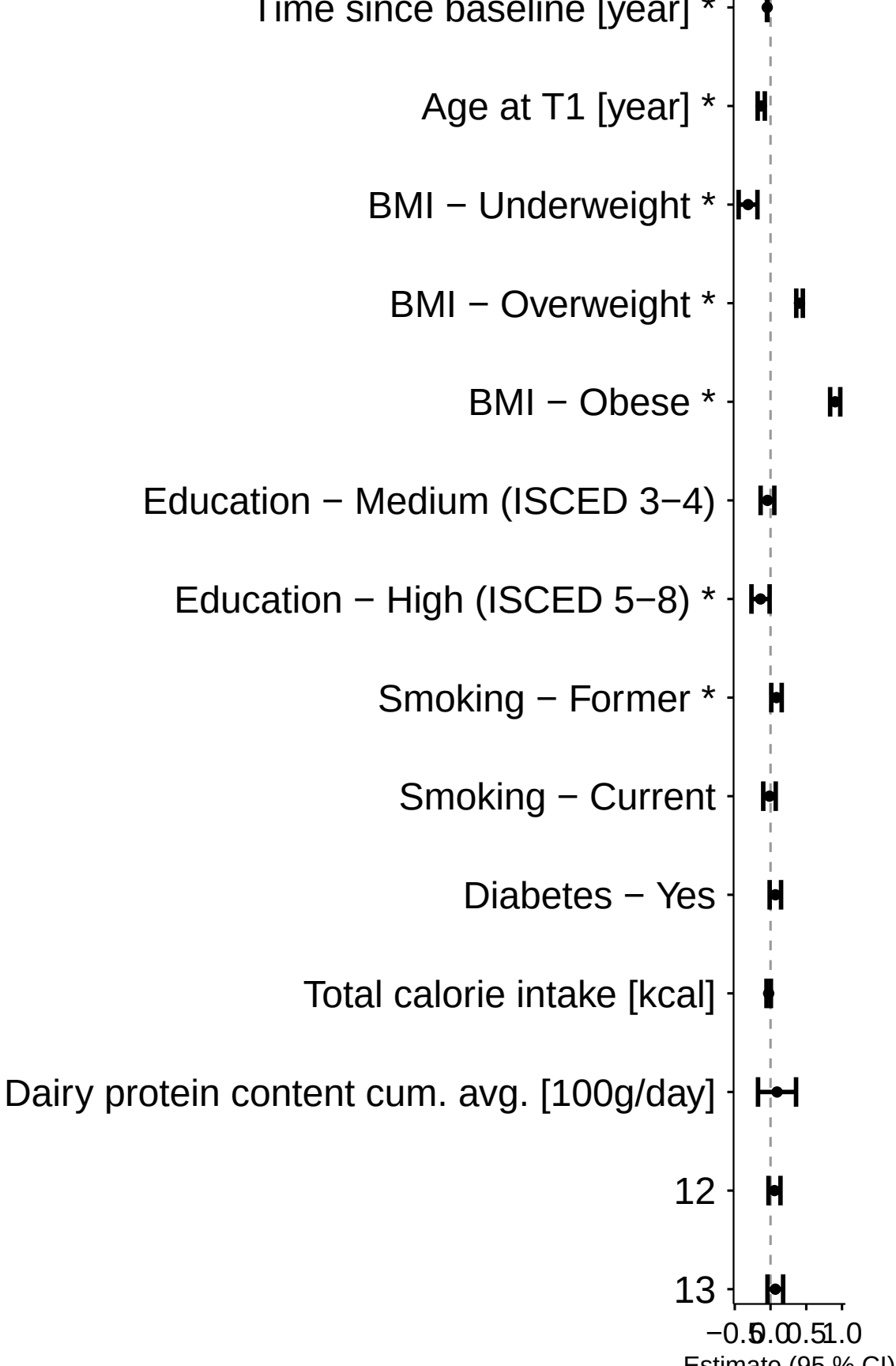
Covariate set: other_PA

age_at_baseline_scaled · BMI_category · education_level · smoking_status · pa_levels_who_f1 · diabetes_status · sumtot1_scaled

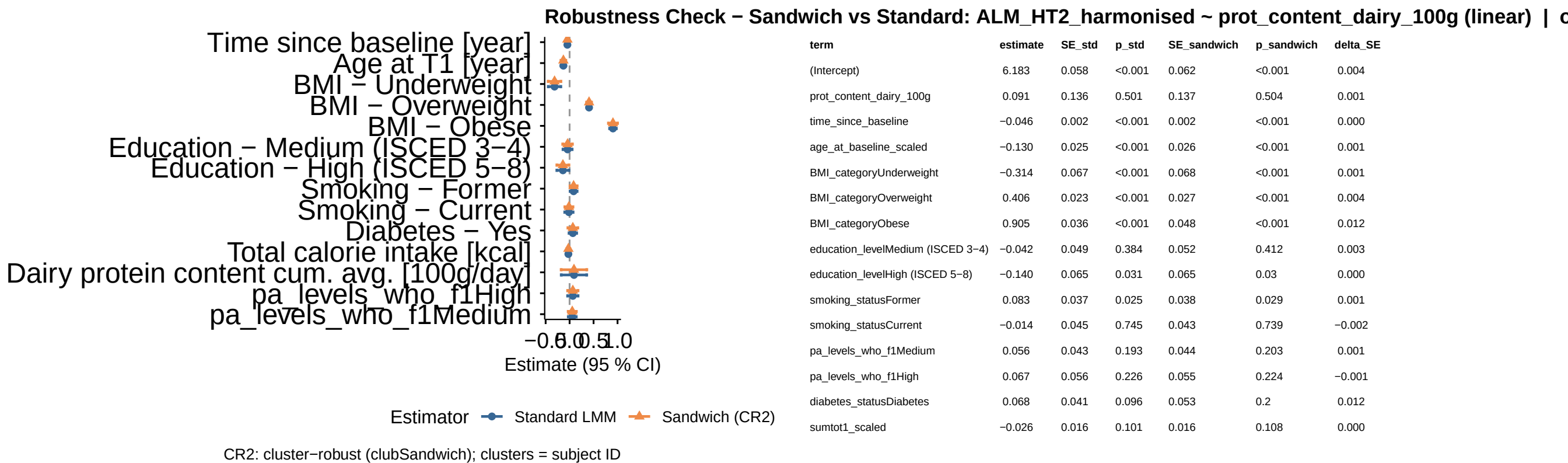
Formula – ALM_HT2_harmonised ~ prot_content_dairy_100g (linear) | other_PA

ALM_HT2_harmonised ~ prot_content_dairy_100g + time_since_baseline +
age_at_baseline_scaled + BMI_category + education_level +
smoking_status + pa_levels_who_f1 + diabetes_status + sumtot1_scaled +
(1 | pt)

Imputations pooled: 10



term	estimate	std_error	statistic	df	p_value	conf_low	conf_high
(Intercept)	6.183	0.058	106.255	2127.1	<0.001	6.069	6.297
prot_content_dairy_100g	0.091	0.136	0.674	746.6	0.501	-0.175	0.357
time_since_baseline	-0.046	0.002	-24.221	100.9	<0.001	-0.050	-0.042
age_at_baseline_scaled	-0.130	0.025	-5.094	2861.6	<0.001	-0.180	-0.080
BMI_categoryUnderweight	-0.314	0.067	-4.688	2489.0	<0.001	-0.445	-0.183
BMI_categoryOverweight	0.406	0.023	17.288	582.5	<0.001	0.360	0.452
BMI_categoryObese	0.905	0.036	25.160	531.9	<0.001	0.834	0.975
education_levelMedium (ISCED 3–4)	-0.042	0.049	-0.870	2890.3	0.384	-0.138	0.053
education_levelHigh (ISCED 5–8)	-0.140	0.065	-2.160	2898.0	0.031	-0.267	-0.013
smoking_statusFormer	0.083	0.037	2.237	2722.9	0.025	0.010	0.157
smoking_statusCurrent	-0.014	0.045	-0.325	2573.6	0.745	-0.102	0.073
pa_levels_who_f1Medium	0.056	0.043	1.303	514.7	0.193	-0.028	0.139
pa_levels_who_f1High	0.067	0.056	1.211	502.1	0.226	-0.042	0.176
diabetes_statusDiabetes	0.068	0.041	1.667	1244.2	0.096	-0.012	0.148
sumtot1_scaled	-0.026	0.016	-1.644	527.6	0.101	-0.056	0.005



Variance Components – ALM_HT2_harmonised ~ prot_content_dairy_100g (linear) | other_PA

```
.. Full VarCorr output .....
Groups   Name          Variance Std.Dev.
pt       (Intercept) 0.315027 0.56127
Residual                0.070425 0.26538

.. Summary .....
Random intercept variance: 0.31503 (SD = 0.56127)
Random slope: not included in this model
Residual variance:         0.07043 (SD = 0.26538)
```

Effect Size (R²) – ALM_HT2_harmonised ~ prot_content_dairy_100g (linear) | other_PA

Marginal R² (fixed effects only): 0.2711

Conditional R² (fixed + random effects): 0.8668

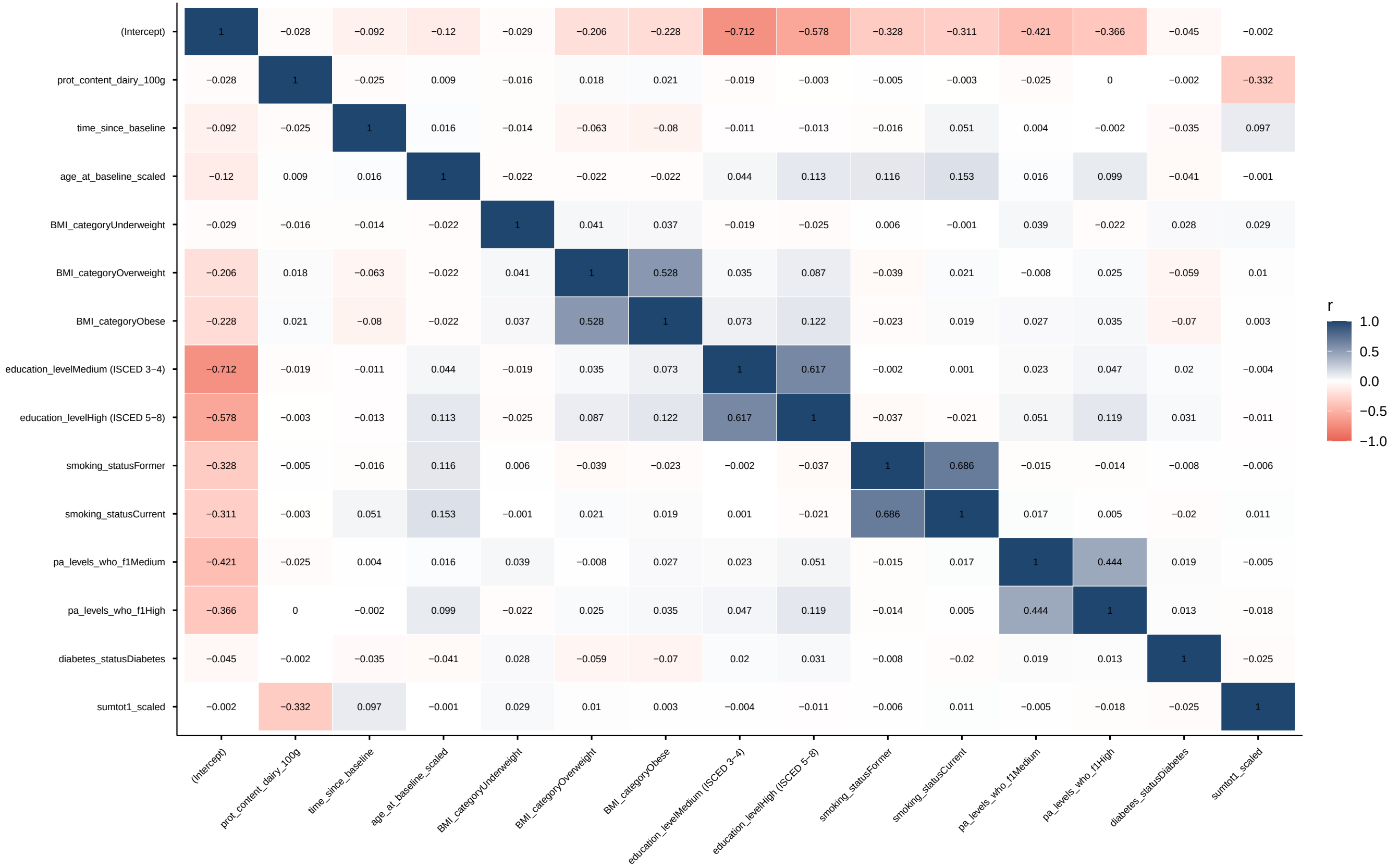
Method: Nakagawa & Schielzeth (2013), Johnson (2014)

VIF – ALM_HT2_harmonised ~ prot_content_dairy_100g (linear) | other_PA

VIF > 5 flagged with '!'

	term	GVIF	Df	$GVIF^{1/(2 \cdot Df)}$	adjusted_vif	flag
	prot_content_dairy_100g	1.126914	1	1.061562	1.062	
	time_since_baseline	1.027188	1	1.013503	1.014	
	age_at_baseline_scaled	1.053614	1	1.026457	1.026	
	BMI_category	1.049476	3	1.008081	1.008	
	education_level	1.050847	2	1.012476	1.012	
	smoking_status	1.045203	2	1.011114	1.011	
	pa_levels_who_f1	1.032882	2	1.008121	1.008	
	diabetes_status	1.013212	1	1.006584	1.007	
	sumtot1_scaled	1.136956	1	1.066281	1.066	

Fixed-Effect Correlations – ALM_HT2_harmonised ~ prot_content_dairy_100g (linear) | other_PA



Influential Points – ALM_HT2_harmonised ~ prot_content_dairy_100g (linear) | other_PA

Influential observations (top 1% Cook's D > 0.29636): 30

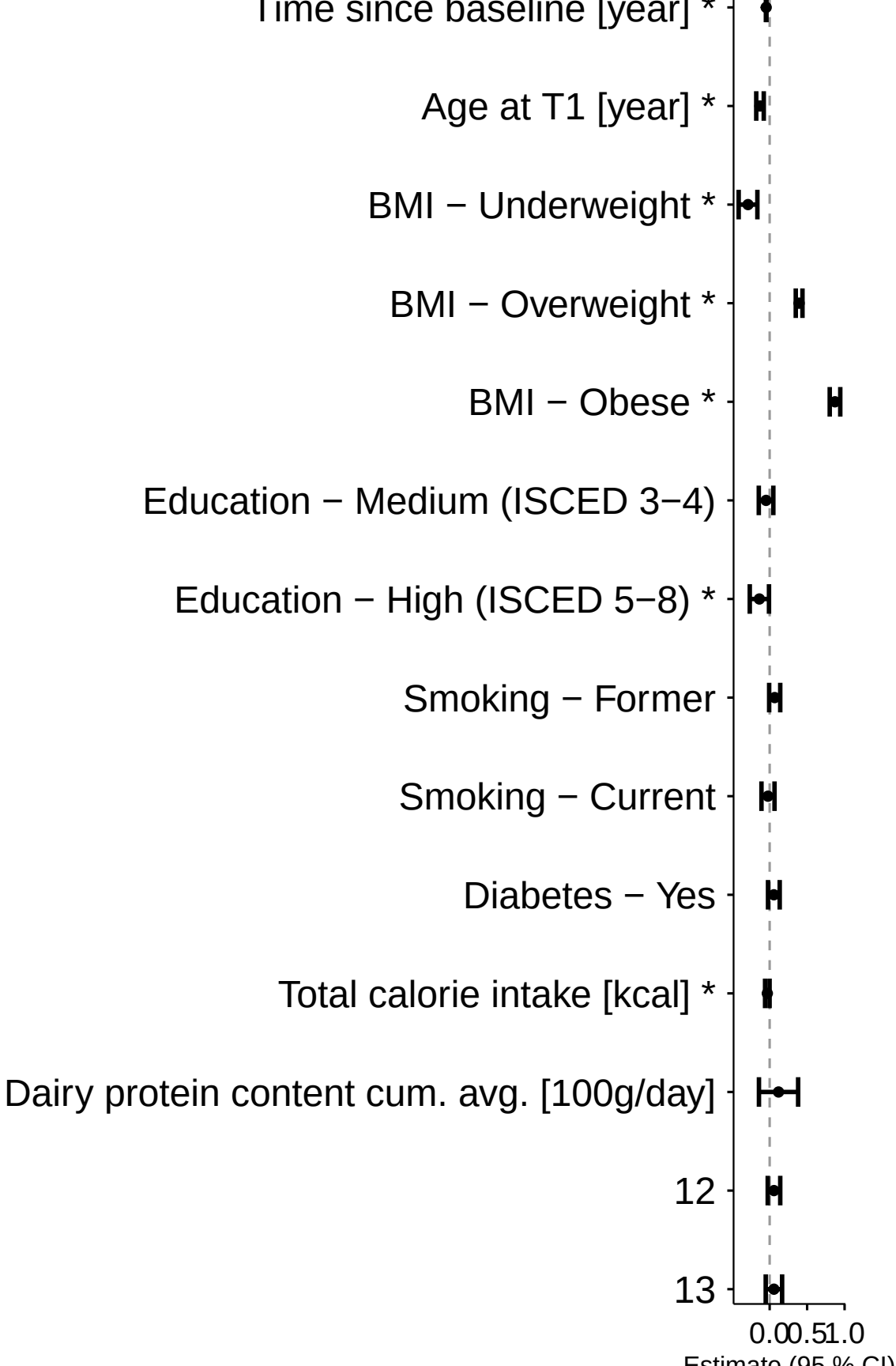
Index	Cook's D
index	cooks_d
172	0.29647
196	0.33426
352	0.34002
353	0.51943
357	0.41966
532	0.30636
724	0.36426
927	0.30203
1061	0.37582
1152	0.54976
1182	0.38209
1218	0.33368
1338	0.32499
1433	0.32257
1544	0.50018
1545	0.80213
1715	0.31172
1717	0.38016
1778	0.47265
1779	0.30190
1785	0.30321
1864	0.48752
2193	0.47068
2194	1.17269
2251	0.56319
2316	0.47792
2317	0.43917
2625	0.35816
2631	0.43140
2632	0.60346

Formula – [Excl. influential (top 1% Cook's D)] ALM_HT2_harmonised ~ prot_content_dairy_100g (linear) | other_PA

Excluded participants (top 1% Cook's D, imp 1): 23

Remaining n: 1043

ALM_HT2_harmonised ~ prot_content_dairy_100g + time_since_baseline +
age_at_baseline_scaled + BMI_category + education_level +
smoking_status + pa_levels_who_f1 + diabetes_status + sumtot1_scaled +
(1 | pt)

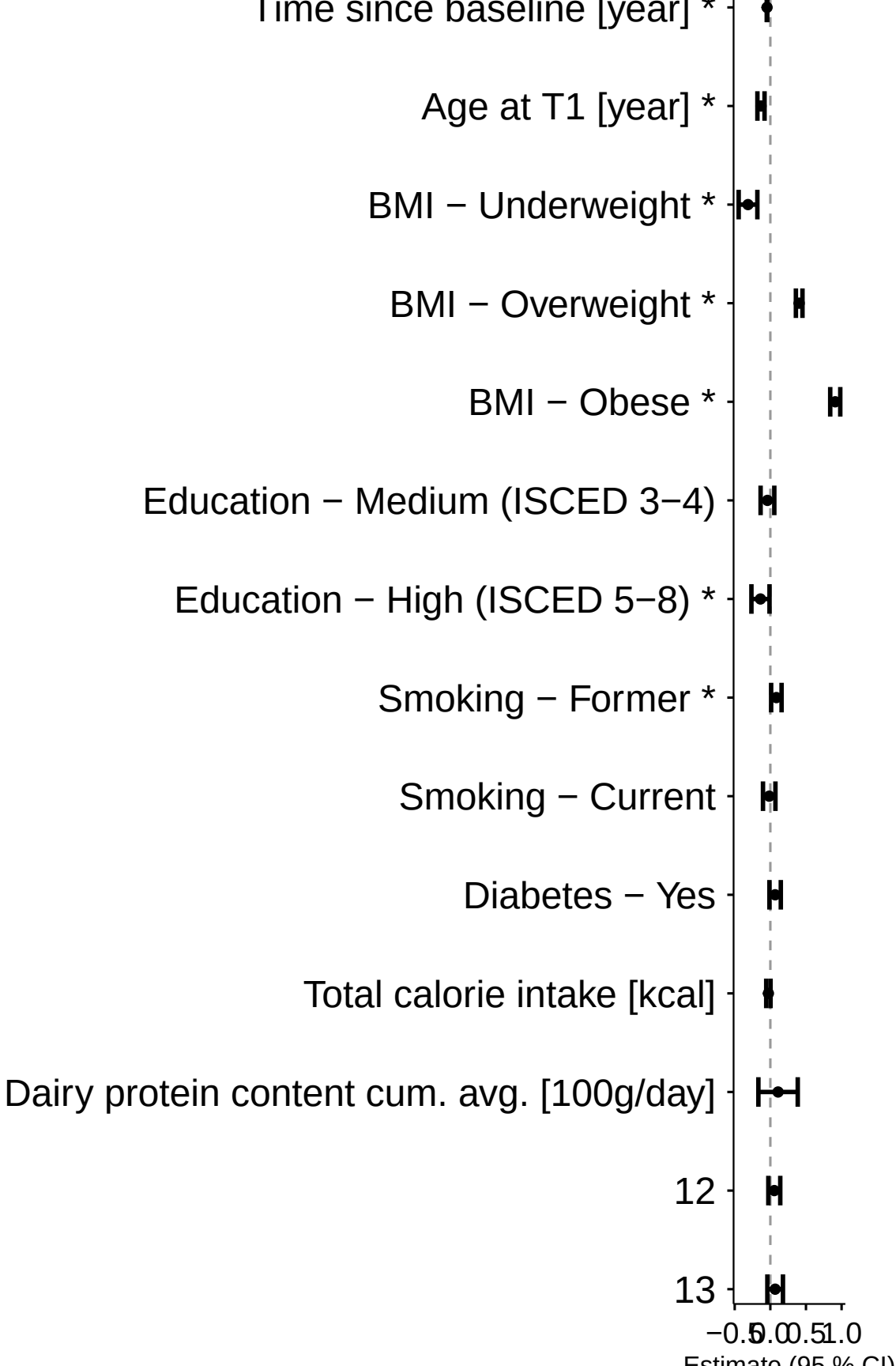


term	estimate	std_error	statistic	df	p_value	conf_low	conf_high
(Intercept)	6.195	0.059	105.844	2180.9	<0.001	6.080	6.310
prot_content_dairy_100g	0.119	0.133	0.891	540.7	0.373	−0.143	0.380
time_since_baseline	−0.047	0.002	−25.048	75.1	<0.001	−0.050	−0.043
age_at_baseline_scaled	−0.129	0.026	−5.031	2820.9	<0.001	−0.179	−0.079
BMI_categoryUnderweight	−0.289	0.064	−4.538	2185.5	<0.001	−0.414	−0.164
BMI_categoryOverweight	0.393	0.023	17.263	446.3	<0.001	0.348	0.438
BMI_categoryObese	0.872	0.036	24.421	363.5	<0.001	0.802	0.942
education_levelMedium (ISCED 3–4)	−0.048	0.049	−0.972	2832.8	0.331	−0.145	0.049
education_levelHigh (ISCED 5–8)	−0.137	0.065	−2.101	2833.1	0.036	−0.265	−0.009
smoking_statusFormer	0.067	0.038	1.782	2553.3	0.075	−0.007	0.141
smoking_statusCurrent	−0.022	0.045	−0.499	2307.6	0.618	−0.109	0.065
pa_levels_who_f1Medium	0.058	0.043	1.353	706.8	0.176	−0.026	0.141
pa_levels_who_f1High	0.057	0.056	1.019	667.2	0.309	−0.053	0.166
diabetes_statusDiabetes	0.056	0.040	1.410	989.7	0.159	−0.022	0.134
sumtot1_scaled	−0.031	0.015	−2.038	253.9	0.043	−0.061	−0.001

Formula – [Sensitivity: dairy <= 1000 g/day] ALM_HT2_harmonised ~ prot_content_dairy_100g (linear) | other_PA

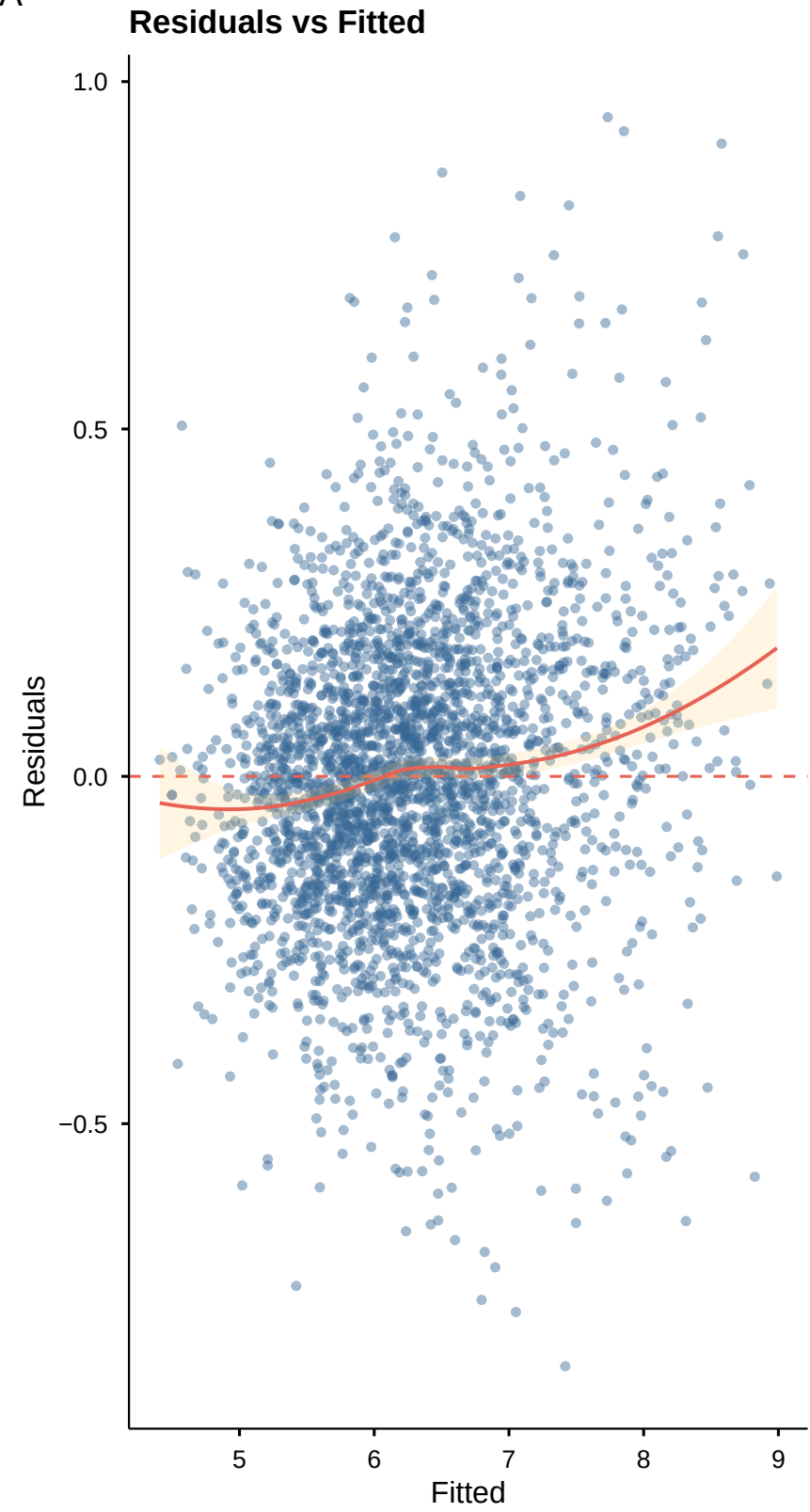
ALM_HT2_harmonised ~ prot_content_dairy_100g + time_since_baseline +
age_at_baseline_scaled + BMI_category + education_level +
smoking_status + pa_levels_who_f1 + diabetes_status + sumtot1_scaled +
(1 | pt)

Imputations pooled: 10
Participants in sensitivity sample: 1061
Threshold: dairy_total_gday_cumavg <= 1000 g/day

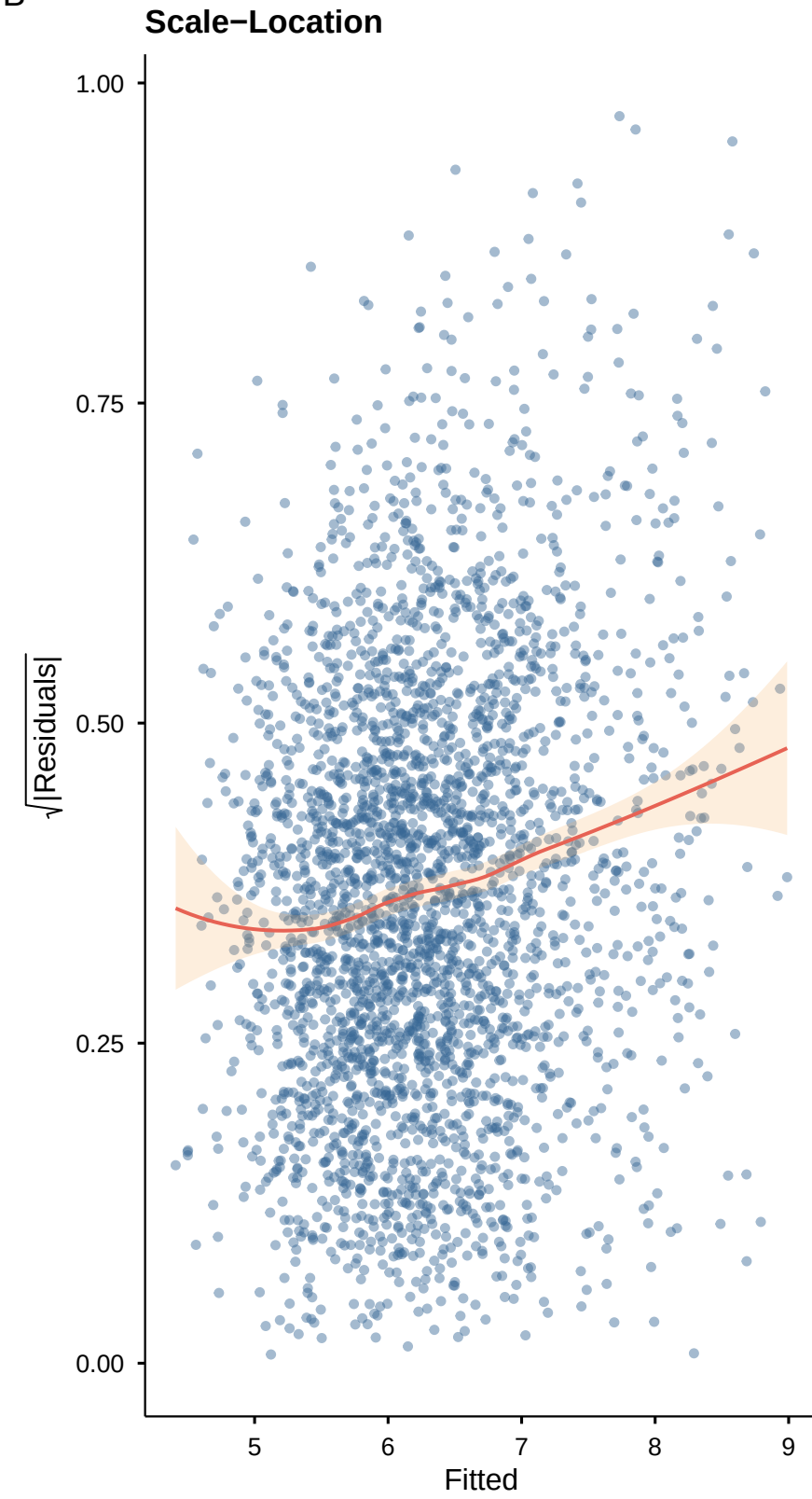


term	estimate	std_error	statistic	df	p_value	conf_low	conf_high
(Intercept)	6.181	0.058	106.013	2180.9	<0.001	6.067	6.296
prot_content_dairy_100g	0.109	0.141	0.772	659.6	0.441	-0.168	0.385
time_since_baseline	-0.046	0.002	-24.162	94.1	<0.001	-0.050	-0.043
age_at_baseline_scaled	-0.131	0.025	-5.153	2858.1	<0.001	-0.181	-0.081
BMI_categoryUnderweight	-0.313	0.067	-4.685	2499.4	<0.001	-0.444	-0.182
BMI_categoryOverweight	0.405	0.023	17.240	605.5	<0.001	0.359	0.451
BMI_categoryObese	0.911	0.036	25.392	600.1	<0.001	0.840	0.981
education_levelMedium (ISCED 3–4)	-0.040	0.049	-0.826	2883.2	0.409	-0.136	0.056
education_levelHigh (ISCED 5–8)	-0.138	0.065	-2.136	2885.1	0.033	-0.265	-0.011
smoking_statusFormer	0.085	0.037	2.263	2697.5	0.024	0.011	0.158
smoking_statusCurrent	-0.016	0.045	-0.352	2500.9	0.725	-0.103	0.072
pa_levels_who_f1Medium	0.056	0.043	1.306	538.9	0.192	-0.028	0.139
pa_levels_who_f1High	0.068	0.056	1.218	521.1	0.224	-0.042	0.177
diabetes_statusDiabetes	0.067	0.041	1.659	1225.2	0.097	-0.012	0.147
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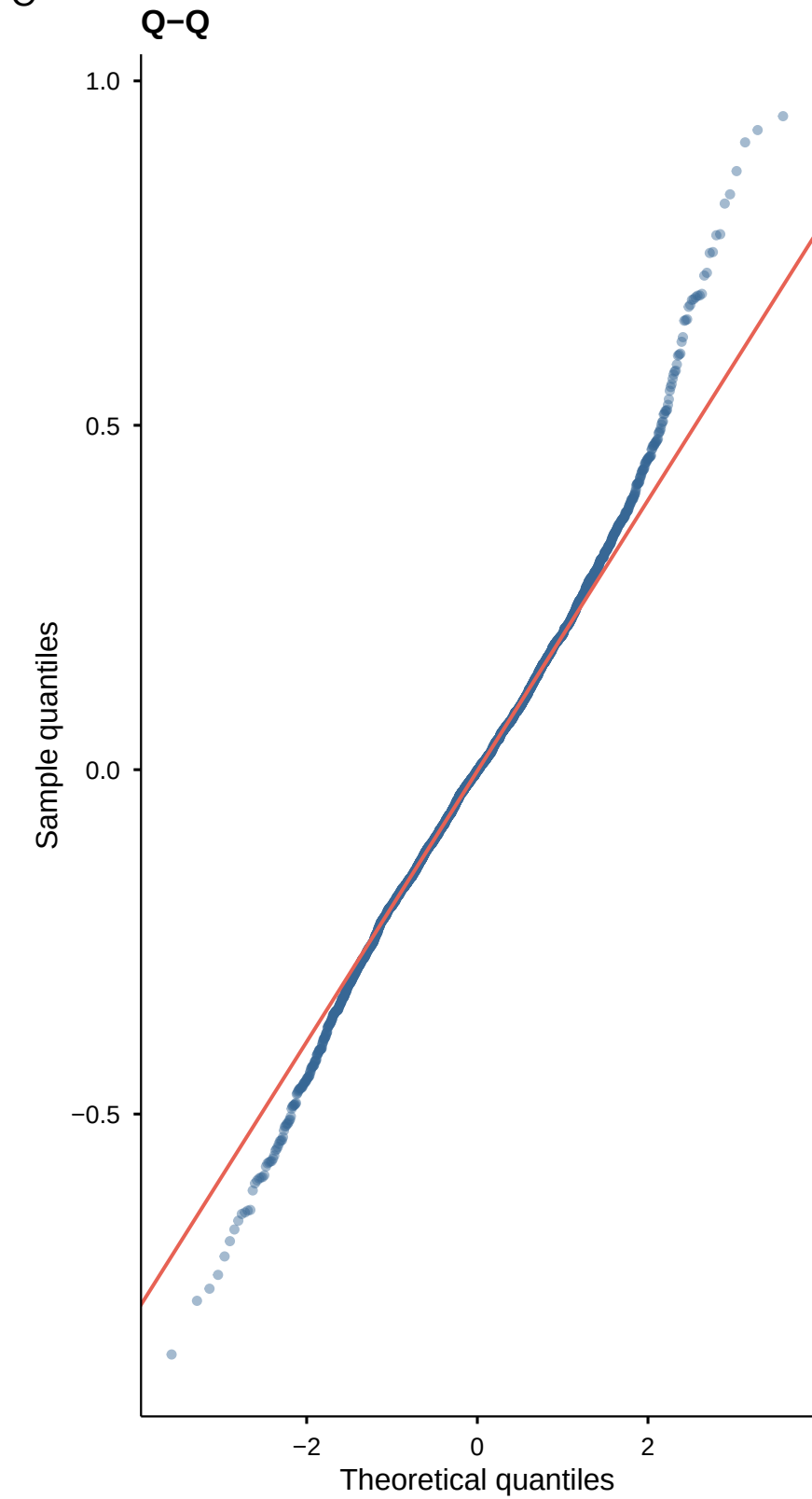
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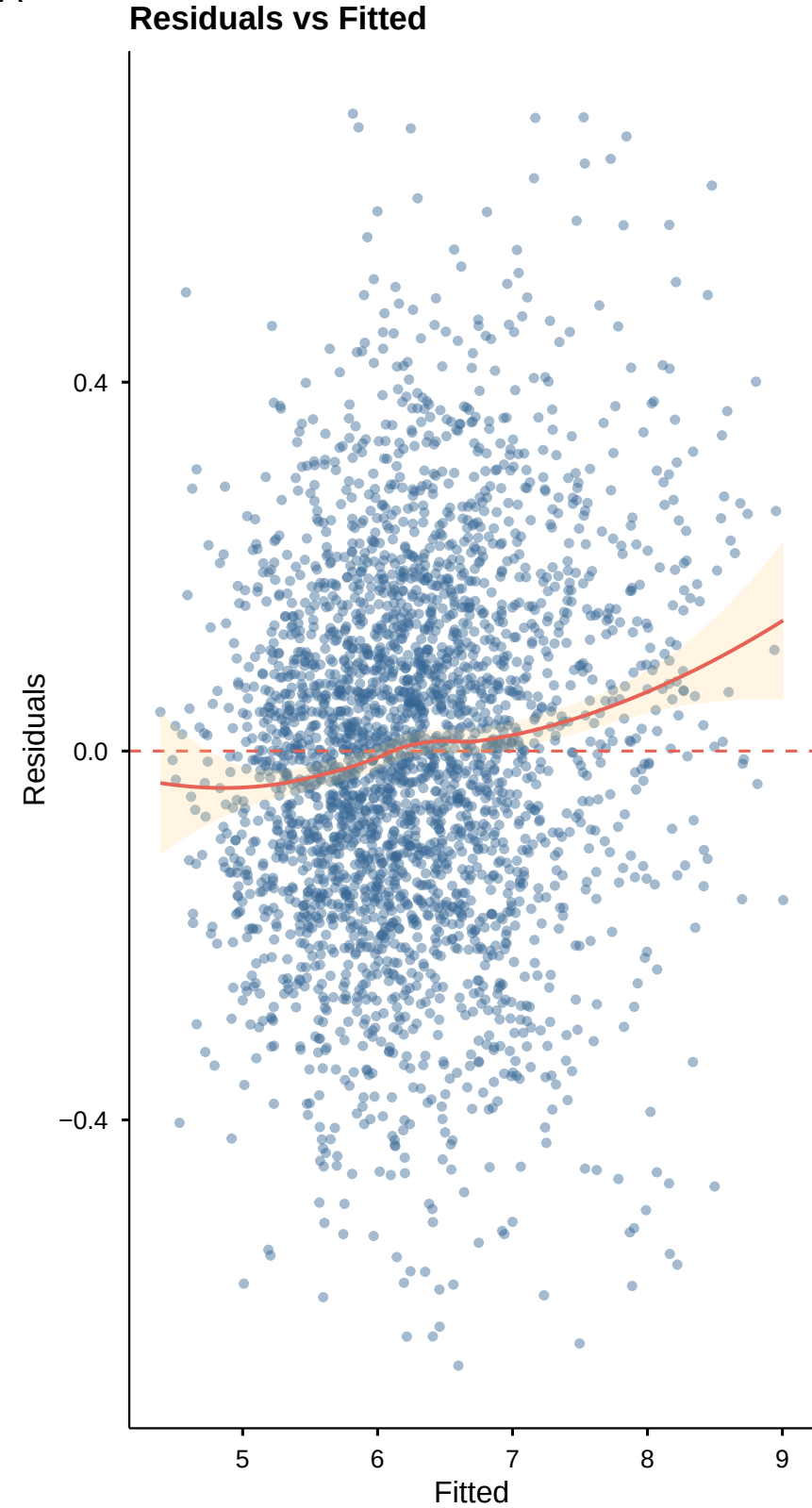
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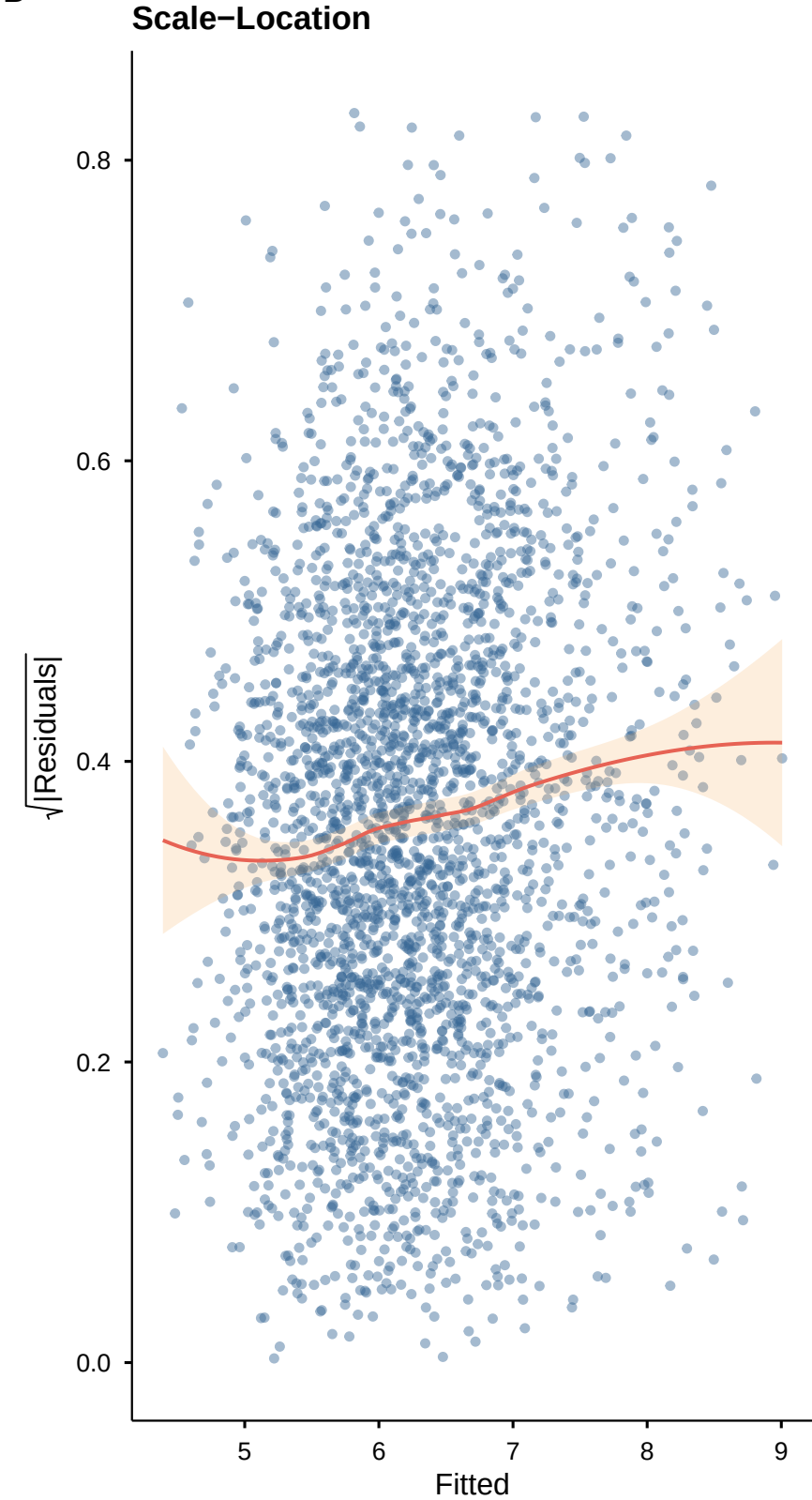
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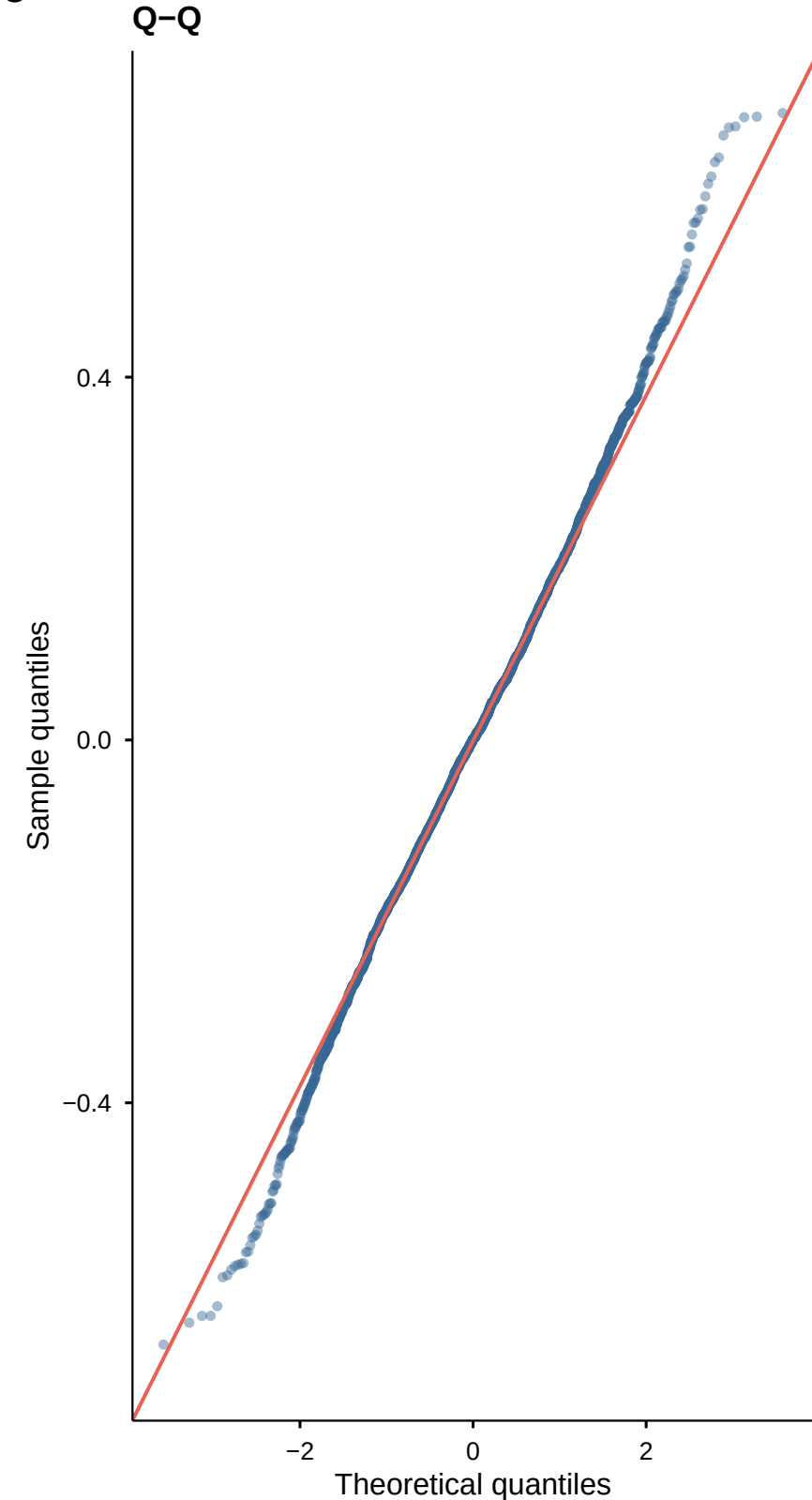
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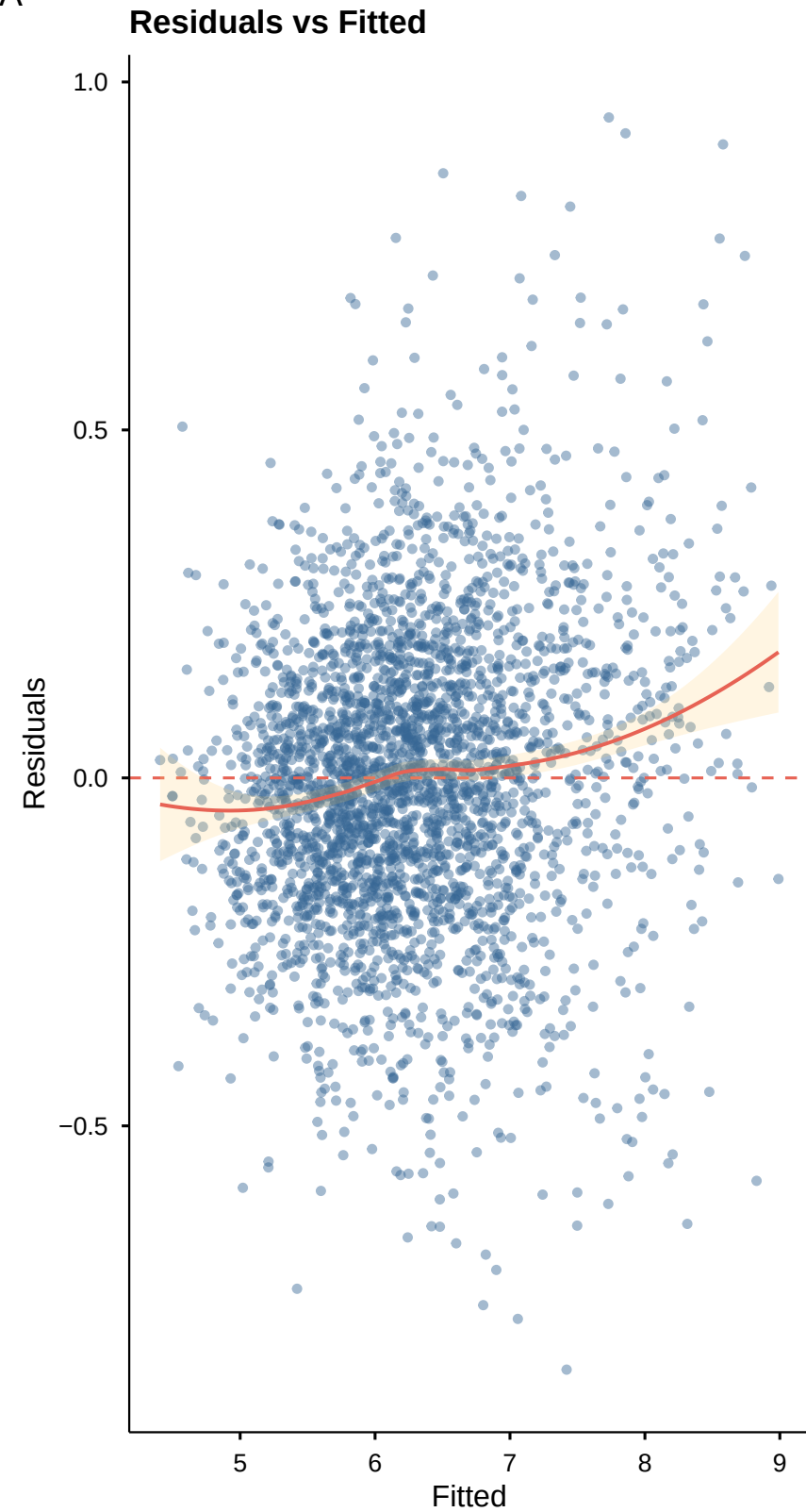
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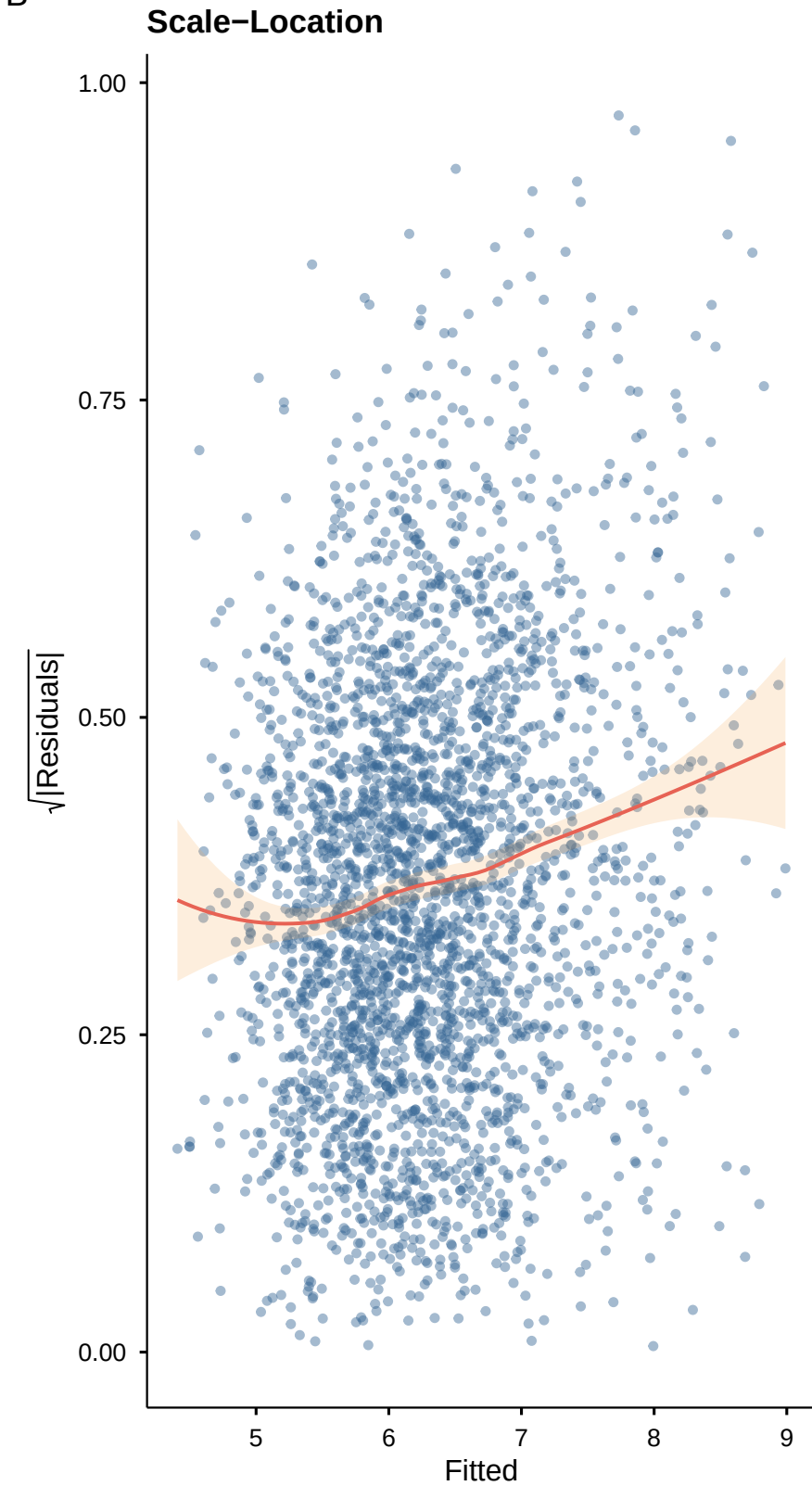
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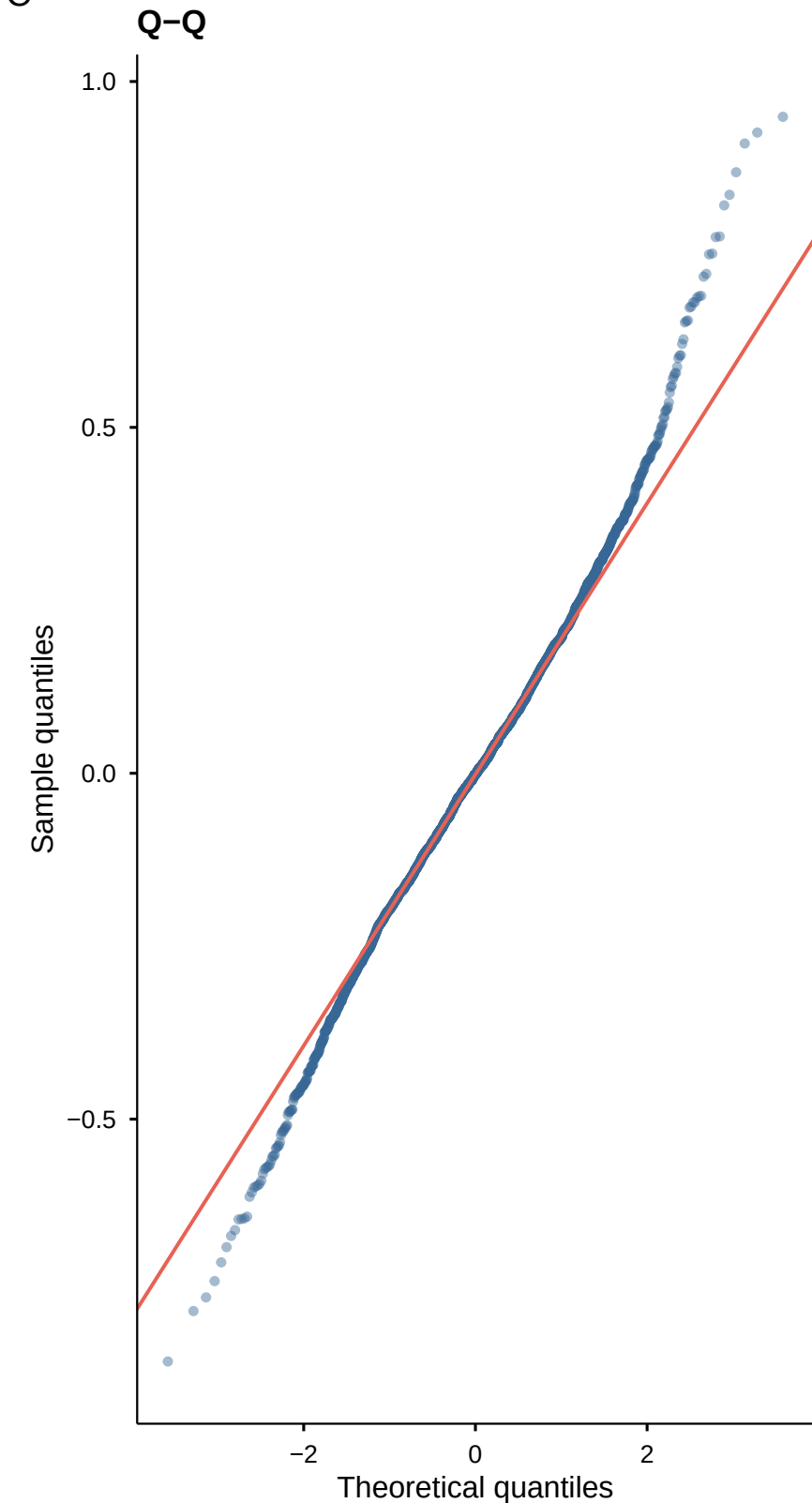
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visNetwork	2.1.4	[90m2025-09-04 [39m [90m[1] [39m [1m [35mRSPM (R 4.5.0) [39m [22m
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[41m [37m* [39m [49m .. Packages attached to the search path.

[41m [37mP [39m [49m .. Loaded and on-disk path mismatch.

[1m [36m..... [39m [22m